

Mode Collapse in RLVR & ModeBench

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Abstract

Correctness alone does not reveal how many successful solutions a model can produce. We introduce MODEBENCH, five executable domains that identify task-defined verified modes. Matched replay comparisons improve average held-out correctness and sampled breadth, with heterogeneous effects and incomplete larger-model and harder-task cohorts. Hosted evaluations also find accurate but concentrated outputs. Categorical analysis explains how binary rewards can amplify common modes and how fixed-bank replay can oppose that pressure under stated assumptions.

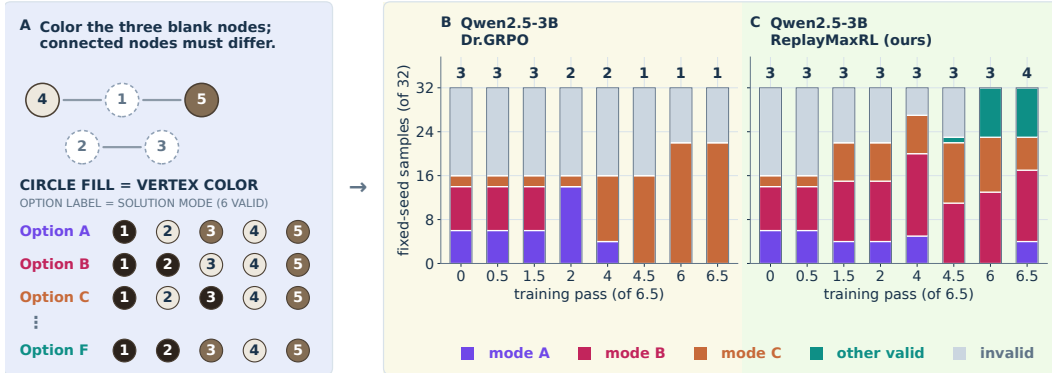


Figure 1: **Correctness can rise while sampled alternatives narrow.** One graph admits six valid colorings. Bars pool four eight-sample draws from matched Qwen2.5-3B runs. At pass 6.5, Dr.GRPO produces 22/32 correct from one mode; ReplayMaxRL produces 32/32 across four. This illustration changes both the fresh objective and replay; Figure 5 separates them.

1 Preserving alternatives for mathematical agents

Mathematical agents can check alternative computations only if the model proposes them [Brown et al., 2024, Snell et al., 2024]. In reinforcement learning with verifiable rewards (RLVR), binary correctness feedback rewards familiar and new successes equally [Guo et al., 2025]. Accuracy can therefore improve while sampled verified outputs narrow (Figure 1).

We store verified outputs by identity and revisit them during training.

2 MODEBENCH makes output identity executable

MODEBENCH assigns each accepted output a canonical key defining its verified *mode* (Figure 2): computation routes in Countdown/MathIR, tested behaviors in Python, and configurations in

18 Graph/PantryPlan. Aliases merge; distinct routes can share an answer (Appendix B). Sampled
 19 modes do not identify total support or latent reasoning diversity.

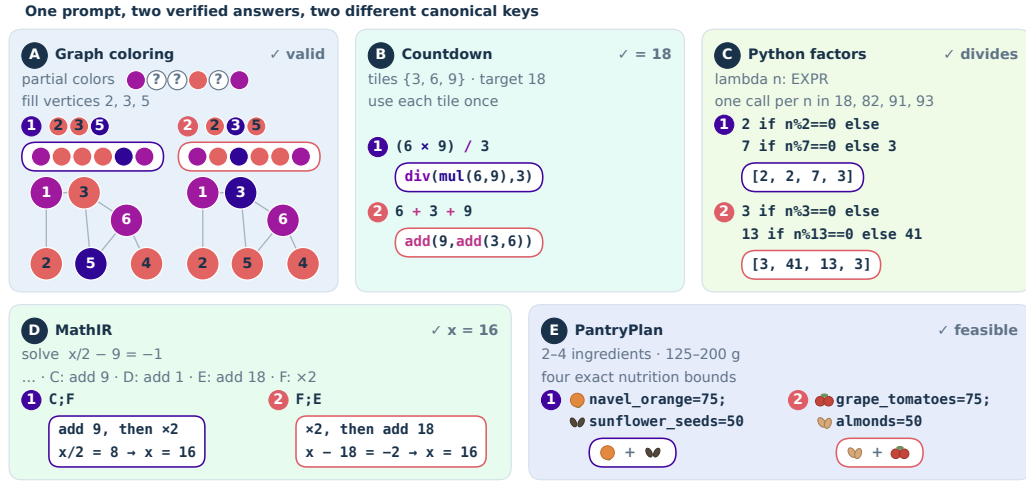


Figure 2: **One prompt, different verified outputs.** Canonical keys identify valid configurations, computation routes and tested behaviors. They merge cosmetic aliases, without identifying latent reasoning.

20 3 ReplayMaxRL: fresh learning with verified memory

21 ReplayMaxRL combines MaxRL’s sampling-aware learning from new samples [Tajwar et al., 2026]
 22 with replay of earlier verified successes (Figure 3).

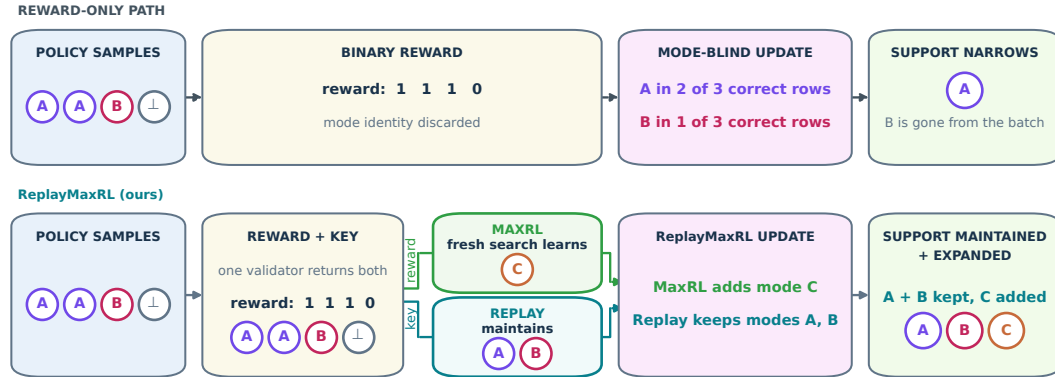


Figure 3: **Discovery and preservation use the same verified samples.** MaxRL learns from fresh rewards; replay revisits discovered modes. Memory cannot protect a mode before it is discovered.

23 For bank $\mathcal{B}_x$ with one policy-generated exemplar e_c per admitted key, replay averages token-
 24 normalized log likelihoods:

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_{\theta}(e_c | x)}{|e_c|}. \quad (1)$$

25 Equal weights revisit keys regardless of discovery frequency. ReplayDr.GRPO uses Dr.GRPO [Liu
 26 et al., 2025] for fresh updates. Fixed-bank replay bounds probabilities away from zero in categorical
 27 mean flows, not neural training or observable sampling (Appendices K.4, D).

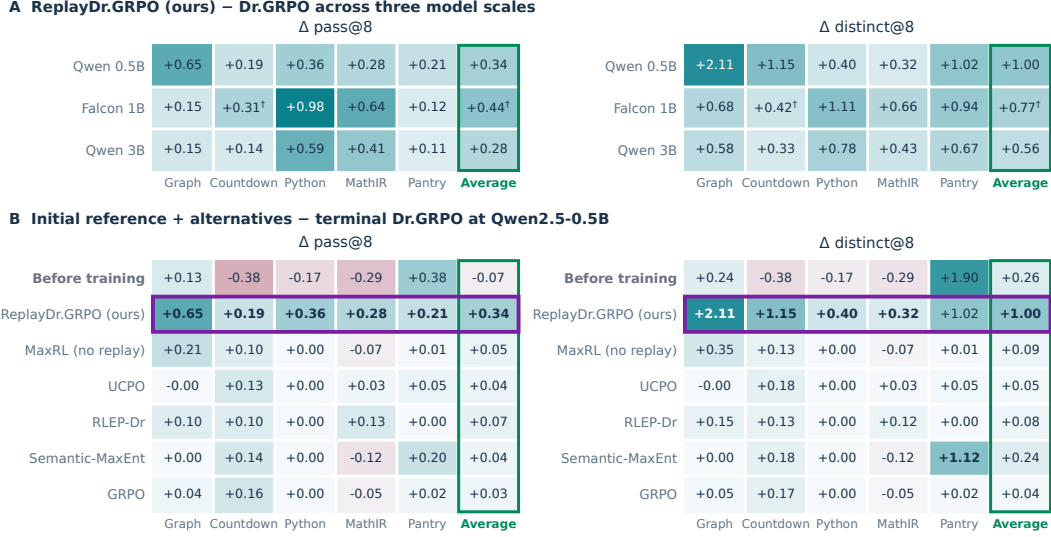


Figure 4: **Replay effects on correctness and sampled breadth across scales.** (A) ReplayDr.GRPO minus Dr.GRPO. (B) Qwen2.5-0.5B alternatives and the initial model minus Dr.GRPO. Daggers indicate four pairs; otherwise five. Averages use shared seeds; intervals: Appendix E. Models: 0.5B, 1B, 3B.

28 4 Does memory preserve verified output support?

29 We report the chance of any correct output (pass@8), mean distinct verified modes (distinct@8),
 30 and their difference, *extra modes* beyond the first success. Counts remain coupled to correctness;
 31 eight samples do not identify total positive-probability support.

32 Five registered seeds train for eight passes. Each domain has 128 held-out prompts with four eight-
 33 sample draws; none enter the bank. Controls use the same bank code with zero replay gradient.
 34 Figure 4 also compares GRPO, UCPO [Lochab et al., 2026], RLEP-Dr success replay, and semantic
 35 entropy (protocols: Appendix B).

36 **Replay improves average correctness and sampled breadth.** ReplayDr.GRPO improves both
 37 domain averages at each scale (Figure 4). Uniform key weighting adds .32 extra modes over frequency
 38 weighting, on average at Qwen2.5-0.5B; its correctness contrast is inconclusive (Appendix E.6).

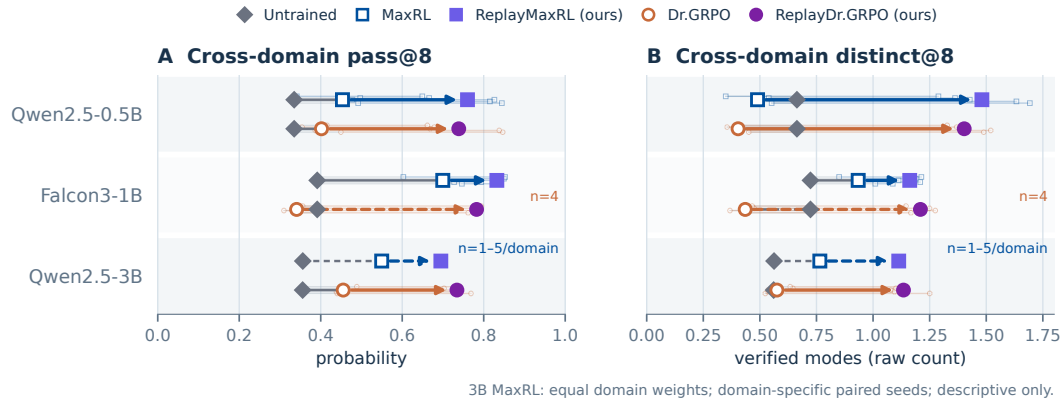


Figure 5: **MaxRL and replay at all three model scales.** Tracks connect initial, trained, and replay models. Complete tracks average within seed; Falcon Dr.GRPO uses four. Qwen2.5-3B MaxRL equally averages five paired domain means ($n = 5, 3, 5, 3, 1$): dashed, descriptive, no interval. Archived models.

39 **Replay complements both fresh objectives on average.** Adding replay improves average correctness
40 and sampled breadth under both objectives at Qwen2.5-0.5B and Falcon3-1B (Figure 5). The
41 Qwen2.5-3B MaxRL summary uses partial seed sets without an interval (Appendix E.2).

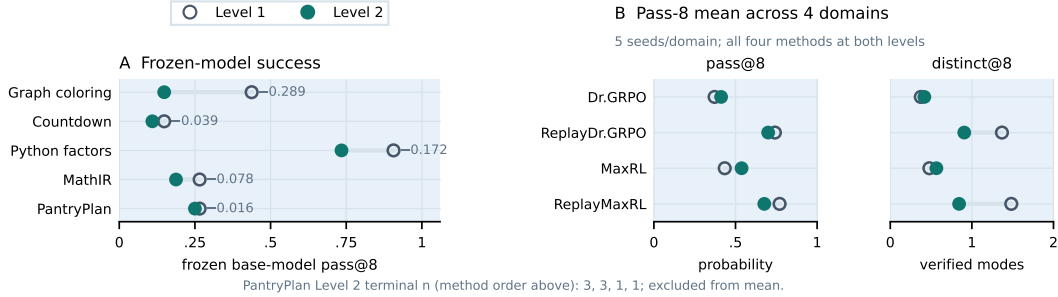


Figure 6: **Replay across difficulty levels with Qwen2.5-0.5B-Instruct.** Open/filled markers: Level 1/2. (A) Frozen-model development admission. (B) Terminal means weight four complete domains equally within five paired seeds, shared across both levels and all methods. Pantry arm counts remain separate (Appendix E.3). Archived models.

42 **Replay gains extend to the harder-level average.** Level 2 retains validators and keys. Its con-
43 struction reserves differ from the native Level-1 terminal tests. Four complete five-seed factorials
44 show positive mean replay effects on correctness and sampled breadth within Level 2 (Figure 6);
45 PantryPlan remains partial. Benefits vary: Countdown adds .017 extra modes under ReplayMaxRL,
46 but its correctness interval includes zero (Appendix E.2).

47 **Accurate hosted outputs can remain concentrated.** Seven deployments vary in breadth across five
48 domains and three levels; accurate Graph and MathIR outputs often revisit few modes (Figure 7).
49 A no-reasoning GPT sweep raises aggregate pass@8 and breadth through $T = 1.5$ (Appendix H.3).
50 Prompt and provider settings differ; these observations identify neither training causes nor model-size
51 effects.

52 **Conclusion and limits.** Replay can improve correctness and sampled breadth; only admitted
53 exemplars receive replay. Fixed-bank scores still have a declining tail (Appendix F.3). Exact mode
54 probabilities, causal accuracy mechanisms and benefits under changed constraints remain unverified.

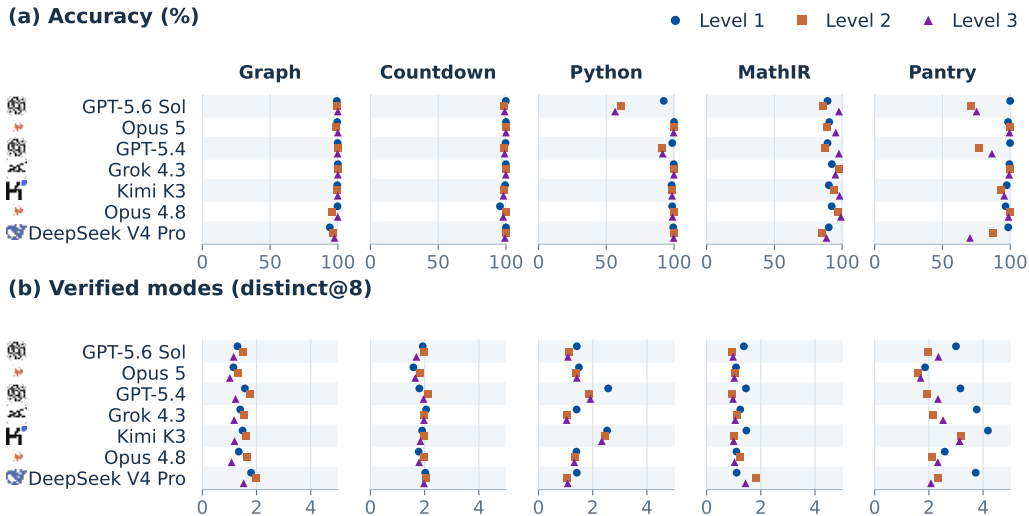


Figure 7: **High accuracy need not imply broad verified outputs.** Each cell uses all eight stateless draws on 128 prompts after frozen formatting normalization. Opus 5 Python uses revised wording without a system message, precluding a common-prompt ranking. Original protocols: Appendix H.

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139 Supplementary Material

140 This supplement gives formal metrics, benchmark and protocol details, the replay algorithm, current
 141 results and training curves, mechanism checks, core proofs, and reproducibility. Model scales and
 142 available paired cohorts match the main figures; detailed histories and theoretical extensions remain
 143 in the research archive.

144 A Formal Metric Definitions

145 For prompt x with specification s_x , the validator maps an output to a canonical valid key or failure:
 146 $V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}$. The learner sees keys only for its generated responses, never the full
 147 catalogue $\mathcal{C}_x$ or its size. For $Y \sim \pi_\theta(\cdot | x)$, define

$$P_\theta^+(x) = \Pr[V(Y, s_x) \neq \perp], \quad p_\theta^+(c | x) = \Pr[V(Y, s_x) = c | V(Y, s_x) \neq \perp].$$

148 The conditional distribution is defined only when $P_\theta^+(x) > 0$; otherwise all sampling metrics below
 149 are zero. Mode collapse means concentration onto fewer valid keys, which can coexist with increasing
 150 success probability.

151 For K independent responses, let

$$D_K = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad F_K = K^{-1} \sum_i \mathbf{1}[V(Y_i, s_x) \neq \perp].$$

152 Then $\text{distinct@}K = \mathbb{E}[D_K]$ counts expected unique verified keys in $[0, K]$, whereas $\text{pass@}K =$
 153 $\Pr[D_K > 0]$ and $\text{mean@}K = \mathbb{E}[F_K]$ are probabilities in $[0, 1]$. The samplewise inequality $F_K \leq$
 154 $\mathbf{1}[D_K > 0] \leq D_K$ gives $\text{mean@}K \leq \text{pass@}K \leq \text{distinct@}K$, but changes in the count and
 155 probability have different units.

156 **Lemma A.1** (Success and sampled breadth). *Fix a prompt, a finite or countable valid-key set, and*
 157 *$K \geq 1$ independent draws from the same fixed sampling law. Let $\mu_c = \Pr[V(Y, s_x) = c]$ and*
 158 *$P = \sum_c \mu_c$. Then*

$$S_K := \text{pass@}K = 1 - (1 - P)^K, \quad B_K := \text{distinct@}K = \sum_c [1 - (1 - \mu_c)^K]. \quad (2)$$

159 *For a finite valid-key set of size $m \geq 1$ and $P > 0$, write $q_c = \mu_c/P$. At fixed P , B_K is Schur-*
 160 *concave in q : if q majorizes q' (its decreasingly sorted partial sums are at least those of q'), then*
 161 *$B_K(P, q) \leq B_K(P, q')$. In particular,*

$$1 - (1 - P)^K \leq B_K(P, q) \leq m[1 - (1 - P/m)^K]. \quad (3)$$

162 *For $K \geq 2$, the lower bound is attained only by concentration on one key, and the upper bound only*
 163 *by uniform q . For $K = 1$, $B_1 = P$ for every q . At fixed q , increasing P cannot decrease B_K .*

164 *Proof.* All K draws fail with probability $(1 - P)^K$. For each key c , its appearance indicator has
 165 expectation $1 - (1 - \mu_c)^K$; summing indicators gives the second identity, also for countably many
 166 keys by nonnegativity. For fixed $P > 0$ and $K \geq 2$, the function $f(v) = 1 - (1 - Pv)^K$ is strictly
 167 concave on $[0, 1]$. Transferring mass from a larger coordinate to a smaller one without reversing
 168 their order increases $\sum_c f(q_c)$; successive such transfers give the majorization assertion. The point
 169 mass and uniform vector are its extreme cases, giving the bounds and equality conditions. Finally,
 170 $\partial B_K / \partial P = K \sum_c q_c (1 - Pq_c)^{K-1} \geq 0$, which proves monotonicity in correctness. $\square$

171 These are promptwise identities, before averaging over prompts. In particular, applying the nonlinear
 172 formula to a mean correctness generally changes the result. They explain the shared measurement in
 173 the training and hosted studies: success depends on total correct mass; sampled breadth also depends
 174 on its allocation. More breadth alone proves neither greater conditional diversity nor better success,
 175 and neither metric establishes downstream utility.

176 Raw $\text{distinct@}K$ measures sampled coverage, not total positive-probability support. Uniform
 177 independent sampling from N finite actions, with n_c aliases of valid key c , gives $\sum_c [1 - (1 - n_c/N)^K]$
 178 expected keys; open-ended program spaces have no canonical uniform generator.

The terminal co-primary outcomes use $K = 8$: success probability `pass@8` and raw `distinct@8`. Their difference $\mathbb{E}[D_K - \mathbf{1}[D_K > 0]]$ counts extra verified modes beyond the first. This interpretive decomposition remains coupled to correctness and is neither a third primary endpoint nor a success-matched diversity measure.

A.1 Why balance within the replay bank

Uniform sampling over keys prevents discovery frequency from determining replay frequency. To explain its relationship to the evaluation metric, we first analyze an equal-weight categorical surrogate, without length normalization. A stored response’s probability need not equal its key’s total probability. For $C = |\mathcal{B}_x|$ banked keys, write $p(c)$ for their probabilities and q_B for their total mass. Then

$$-\frac{1}{C} \sum_{c \in \mathcal{B}_x} \log p(c) = -\log q_B + \log C + \text{KL}(U_C \parallel p(c \mid c \in \mathcal{B}_x)). \quad (4)$$

The objective rewards greater banked verified mass and a more balanced conditional distribution, aligning with `pass@K` and `distinct@K`. The combined update need not improve both at every step. Indeed banked-key coverage contributes $\sum_c [1 - (1 - p(c))^K]$ to expected `distinct@K`; at fixed q_B and C , concavity makes it largest under uniform allocation, while $1 - (1 - q_B)^K$, the probability of any banked key, depends only on total mass. This deliberate objective-metric alignment motivates treating held-out `pass@K` as co-primary. Uniformity also maximizes the smallest conditional banked-key mass, making equal retention explicit. Banks contain only training-prompt responses; evaluation responses never enter training.

B Benchmark and Evaluation Protocol

MODEBENCH defines correctness and identity through the same executable validator: invalid outputs receive reward zero and no key; valid outputs receive reward one and a canonical execution key. Neither the complete valid-key catalogue nor its size is supplied to training, replay, scheduling, or stopping. Table 1 specifies the five domains and which surface forms merge. The public dataset preserves each domain’s split roles and construction identity. This paper’s training comparisons use Levels 1 and 2; the separate Level-3 release does not imply completed Level-3 training results.

Table 1: **MODEBENCH**. Correctness and identity come from one executable path; training starts without the mode catalogue. *Valid modes* is the mean number of correct execution modes per evaluation prompt, exact over the released multi-answer split. Every domain trains on 384 prompts and evaluates on a fixed 128-prompt split; the exact prompt each policy receives is reproduced verbatim in Appendix C for a direct, byte-level audit of the evaluation interface.

Domain	Generated answer / check	Executed key	Aliases that merge	valid modes
Graph coloring	Complete assignment; check fixed colors and edges	Color vector	Spacing only	6.27
Countdown	Parse operands; execute exactly	Normalized AST	Commutative order	4.52
Python factors	Execute restricted lambda externally	Return vector	Source programs with same behavior	229.44
MathIR	Execute bounded action program	State trajectory	Programs with the same states	5.00
PantryPlan	Exact ingredient allocation; check all bounds	Ingredient support	Quantity and ordering aliases	18.19

B.1 Split identities and executable validation

Every domain uses 384 training and 128 disjoint evaluation prompts. Python materialization independently executes at least two functions with different valid keys for every row; MathIR executes every certified route through the same interpreter used for model outputs.

Table 2: Deterministic materialization identities. Hashes bind the released rows to their executable checks; the full hashes and construction records accompany the data.

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

PantryPlan. The deterministic v2 release has 384/64/128 training/development/evaluation prompts across four formulation families. Exact support ranges from 8 to 45; the 128 evaluation prompts contain 2,328 feasible supports, averaging 18.1875. Training and evaluation row hashes are a4883c5e...a757f and 5d81ba8d...5923. The validator checks exact inventory, serving steps, mass, nutrient bounds, and dietary tags; identity is ingredient support. A verified allocation is not a claim of palatability, cookability, or safety.

B.2 Python execution contract

Candidates are stripped, limited to 240 characters and 64 AST nodes, and parsed in expression mode. The only accepted form is `lambda n: EXPR` with exactly one ordinary argument named `n`: no defaults, positional-only, keyword, or variadic arguments. Integer literals have absolute value at most 10,000. The expression grammar permits `n`, `+`, `-`, `*`, floor division, remainder, integer comparisons, Boolean and/or, unary plus/minus/not, and conditionals. Boolean and non-integer literals, calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and other identifiers are rejected. Task specifications contain two to eight distinct integers in $[4, 1000]$ and at least two valid behavior vectors.

External execution. The trainer never executes candidate code. A locked parent client launches an isolated Python `-I` process in a fresh session, with a persistent JSON-lines worker accepting only restricted source and trusted task JSON. The worker repeats the AST checks, compiles the expression with empty `__builtins__`, and executes the frozen inputs under a 0.25-second real-time alarm; the parent enforces a separate one-second response deadline. A valid response contains the integer vector and key. Timeouts, empty or malformed responses, broken pipes, non-integer outputs, and failed divisor checks yield $\perp$ and zero reward. Broken pipes, operating-system errors, and malformed responses permit one clean worker restart; a timed-out request remains failed. Worker stdout is never parsed as a candidate answer, and exceptions never produce a key.

Behavioral identity. For every frozen input n_j , the returned d_j must be an integer but not a Boolean, satisfy $1 < d_j < n_j$, and divide n_j exactly. All checks must pass. The key is the ordered vector `python_factor:  $d_1, \dots, d_m$` ; programs merge exactly when their vectors agree. The exact support count $\prod_j |\{d : 1 < d < n_j, d \mid n_j\}|$ is used only for dataset validation.

B.3 Matched Level-2 construction

Level 2 changes problem structure while preserving the verifier and canonicalizer. Its 384/128/128 training/development/evaluation rows have disjoint identities across the Level-1 and Level-2 splits. Training histograms match the native Level-1 training sets. Development and evaluation histograms match designated Level-1 development and confirmation reserves in Graph, Countdown, Python, and MathIR; Pantry uses native references, with its 64-row development histogram doubled to 128 rows.

These construction reserves are not the native Level-1 terminal test sets. The displayed terminal evaluations have different valid-key histograms between levels in Graph, Countdown, and Python. Cross-level changes therefore combine task structure and test population; they do not isolate difficulty at fixed test-set support. Replay comparisons remain paired within each level on the same prompts, seeds, fresh objective, and training budget.

Domain	Level 1	Level 2 structural change
Graph coloring	Three hidden vertices	Four hidden vertices
Countdown	Three operands	Four operands, with values up to 14
Python factors	Four executed cases, all at most 96	Four cases, including a value above 96
MathIR	One-sided linear families	Variable-on-both-sides families
PantryPlan	Base dietary/composition constraints	More active dietary exclusions and tighter composition

Before treatment training, frozen Qwen2.5-0.5B development evaluation admitted a Level-2 domain only when `pass@8` lay in $[.10, .90]$ and was strictly below its construction-reference Level-1 value under identical decoding. The matched-level identity manifest records support histograms, row hashes, construction seeds, disjointness checks, and these admission receipts.

B.4 Training and evaluation settings

Table 3: **Matched run contract.** Settings and systems changes are shared by methods within each model–domain–seed comparison, including the zero-weight replay control.

Component	Setting
Data	384 training and 128 disjoint evaluation prompts per domain and level; no evaluation row enters training or replay.
Fresh rollouts	$G = 16$, temperature 1, top- $p = 1$; one prompt group per update.
Optimization	Eight passes (3,072 updates), one PPO epoch, $\beta_{KL} = 0$; paired initialization, prompt stream, optimizer, hardware class, and checkpoint cadence.
Evaluation	Passes 0, .5, . . . , 8; four fixed temperature-1, top- $p = 1$, $K = 8$ draws per prompt, plus a separate temperature-0 greedy trace. No best-checkpoint selection.
Replay	Capacity $M = 16$ per prompt; one round-robin bank per update; $\lambda_{\text{replay}} = .10$. Controls execute the same code with an exact-zero replay derivative; realized banks and workloads can differ.
Training seeds	Qwen2.5-0.5B: 43–47; Falcon3-1B: 55–59; Qwen2.5-3B: 70–74.
Learning rate	Qwen-0.5B and Falcon-1B: constant 2×10^{-7} , with optimizer-state offload for Falcon. Qwen-3B: cosine decay with 10^{-7} peak, 10^{-8} floor, and 10% warmup.
Response tokens	Graph and Countdown: 192; Python: 192/512/192 at 0.5B/1B/3B; MathIR: 64/128/64; PantryPlan: 8.
Evaluation seeds	Base 610100/610200/610300/610400/76299 for Graph/Countdown/Python/MathIR/PantryPlan; draw d uses base plus d .

Base model	Pinned revision
Qwen2.5-0.5B-Instruct	7ae557604adf67be50417f59c2c2f167def9a775
Falcon3-1B-Instruct	28ba2251970a01dd1edc7ba7dad2eb71216ccfd
Qwen2.5-3B-Instruct	aa8e72537993ba99e69dfaafa59ed015b17504d1

Responses are independent within each evaluation draw. For each training seed, metrics average over all 128 prompts and four draws; raw responses and draw-level records are retained. The co-primary terminal outcomes are `pass@8` and raw `distinct@8`; `mean@8` is diagnostic. Evaluation never supplies training data, bank entries, checkpoint selection, or a stopping rule. Complete paired five-seed blocks receive nominal pointwise 95% Student- t intervals; partial primary intersections report exact counts and descriptive means without intervals. Intervals condition on the frozen prompts and evaluation requests. Domain averages are explicitly labeled, and no missing terminal result is replaced by a shallower checkpoint. Source admission and curve gaps follow Appendix L.1; comparison-specific intersections and the replay-weighting bootstrap are specified with their results.

C Exact Domain Prompts

Each domain uses one system message and one user message, with no few-shot examples. The same template renders training and evaluation for every method. These are the first rows of the frozen evaluation splits, including the exported chat-template wrapper. The checked generator ops/check_

266 paper_domain_prompts.py reproduces them from the datasets. A two-space continuation indent
 267 marks display wrapping: joining it to the previous line with one space recovers the original string.
 268 The PantryPlan listing retains its real newlines.

269 C.1 Graph coloring

270 Template qwen_boxed; 384 training and 128 evaluation
 271 rows in var/data/graph_coloring_modebench_v2.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3), (3,5),
(3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no edge has the
same color on both endpoints. A partial coloring is 2??1?2, where ? means uncolored.
Keep every shown digit fixed. There are exactly 3 question marks. Return exactly one
final answer and no explanation: exactly 3 digits, each one 1, 2, or 3, for the missing
positions from left to right, inside \boxed{<|im_end|>
<|im_start|>assistant
```

272 C.2 Countdown

273 Template qwen_boxed; 384 training and 128 evaluation
 274 rows in var/data/exact_countdown_easy3_probe.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use each
given number exactly once. You may use +, -, *, /, and parentheses. Return exactly one
final answer and no explanation: the expression inside \boxed{<|im_end|>
<|im_start|>assistant
```

275 C.3 Python factors

276 Template qwen_boxed; 384 training and 128 evaluation
 277 rows in var/data/python_factor_modebench_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external Python tool
will call it once for each n in [18, 82, 91, 93]. For every call, return an integer d
satisfying 1 < d < n and n % d == 0. Different valid return vectors are different
solution modes. You may use integer literals, n, +, -, *, //, %, comparisons, Boolean
operators, and conditional expressions; calls, imports, attributes, containers, and
other names are forbidden. Return exactly the one-line lambda inside \boxed{<|im_end|>
<|im_start|>assistant
```

278 C.4 MathIR action menu

279 Template qwen_boxed; 384 training and 128 evaluation
 280 rows in var/data/mathir_action_menu_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed by
exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c. Action
menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E means
sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations, repeated
states, and paths that do not finish with x isolated are rejected. Return only the IDs
separated by semicolons inside the answer box; for example formatting only: \boxed{A;C}.
Do not return x's numeric value.<|im_end|>
<|im_start|>assistant
```

281 C.5 PantryPlan

282 Template qwen_pantry_support_mask; 384 training and 128 evaluation
 283 rows in var/data/pantry_plan_modebench_v2.

```

<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the corresponding
Pantry row in printed order. A trusted environment will choose exact quantities on
precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal, protein=21.5g,
  fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal, protein=0.74g,
  fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal, protein=0.941g,
  fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant

```

284 D Replay Algorithm

285 For each prompt x , $\mathcal{B}_x$ stores at most M verified policy exemplars, one per canonical key; the
 286 serialized cursor τ selects banks for replay. The same update supports either fresh objective.

287 D.1 Fresh task advantages

288 For a group of G fresh responses with verified binary rewards r_i and $m = \sum_i r_i$, Dr.GRPO uses
 289 $A_i = r_i - m/G$, while MaxRL uses

$$A_i^{\text{MaxRL}} = \begin{cases} Gr_i/m - 1, & m > 0, \\ 0, & m = 0. \end{cases}$$

290 The centered MaxRL estimator [Tajwar et al., 2026] gives failures advantage -1 in mixed groups
 291 and weights each success more strongly when successes are scarce. All-correct and all-failure groups
 292 have zero task advantage. Lemma K.2 gives its harmonic pass@ k mean objective. These advantages
 293 apply only to fresh responses; bank exemplars enter the separate verified-likelihood loss.

294 D.2 State and optimizer update

Algorithm 1 One ReplayMaxRL optimizer update

Require: Policy π_θ ; prompt x and verifier specification s_x ; group size G ; fresh objective $o \in \{\text{MaxRL}, \text{Dr.GRPO}\}$; replay weight λ_{replay} ; states $(\mathcal{B}, \tau)$
Ensure: Atomically updated $(\theta', \mathcal{B}', \tau')$

- 1: Draw the on-policy group $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$
- 2: **for** $i = 1, \dots, G$ **do**
- 3: Execute $\tilde{c}_i \leftarrow V(y_i, s_x)$ and observe task reward r_i
- 4: Set $c_i \leftarrow \tilde{c}_i$ iff $r_i > 0 \wedge \tilde{c}_i \neq \perp$; otherwise set $c_i \leftarrow \perp$
- 5: **end for**
- 6: Compute fresh task advantages A_i^o from $\{r_j\}_{j=1}^G$; for ReplayMaxRL use the formula in Appendix D.1
- 7: After the full group is validated, deterministically retain one policy exemplar for each new $c_i \neq \perp$ in $\mathcal{B}_x$, up to capacity M
- 8: $(z, \tau') \leftarrow \text{NEXTBANK}(\{\mathcal{B}_u\}_u, \tau)$ by a fixed round-robin cursor
- 9: **if** a replay bank $\mathcal{B}_z$ exists **then**
- 10: For each retained exemplar e_c , compute $s_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$
- 11: $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} s_c$
- 12: **else**
- 13: $L_{\text{replay}} \leftarrow 0$
- 14: **end if**
- 15: $w_{\text{rep}} \leftarrow (G - 1)/G^2$ for one replay pseudo-group per G on-policy rows
- 16: $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^o) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$
- 17: Checkpoint $(\theta', \mathcal{B}', \tau')$ as one atomic state

295 D.3 Admission, capacity, and deterministic scheduling

296 A bank is keyed by the SHA-256 digest of the exact prompt-token sequence. Only loss-active rows
 297 with positive reward and a nonempty validator key qualify. The full group is validated against the
 298 pre-group bank before any new key is committed. For each new key, the lexicographically smallest
 299 normalized response-token tuple is retained; new keys are admitted in sorted-key order. Admission is
 300 therefore invariant to rollout-row order. The bank keeps one exemplar per key, up to $M = 16$, with
 301 no eviction or replacement. Later keys may enter diagnostics but not replay support when capacity is
 302 full. Prompt-hash collisions or changed tokens under an existing hash are hard errors.

303 At each update, the scheduler sorts nonempty prompt-bank hashes and selects one by a fixed
 304 round-robin cursor; singleton banks are eligible. The cursor is independent of reward, evaluation,
 305 desired support, or entropy, and is serialized with the bank. Stored prompt and response tokens are
 306 materialized unchanged into a right-padded teacher-forcing batch. Prompt and padding positions
 307 are masked. Each exemplar contributes its mean response-token log probability, averaged uniformly
 308 across keys in the selected bank before applying λ_{replay} and the exact $(G - 1)/G^2$ rollout scaling in
 309 Algorithm 1. Neither exhaustive support nor evaluation outputs enter this loss.

310 D.4 Matched controls and recoverable state

311 Controls share validation, bank updates, scheduling, materialization, and the score pass, with the
 312 replay score-gradient vector replaced by exact zeros. Treatment activates only that derivative. Policies
 313 can subsequently differ in samples, bank contents, selected banks, exemplar lengths, and realized
 314 computation. Logs retain bank/key counts, replayed token counts, raw and applied losses, score-
 315 gradient summaries, cursor phase, and the exact-zero switch. Model and optimizer state, bank counts
 316 and exemplar tokens, replay and data cursors, and request-stream state are checkpointed atomically;
 317 the loader validates schema, capacity, scheduler and estimator constants, token types, prompt hashes,
 318 key/exemplar consistency, and cursor bounds before accepting a resume.

319 D.5 Fixed Semantic MaxEnt comparator

320 Fixed Semantic MaxEnt is an entropy-inspired fresh-rollout comparator. For validator-positive
 321 canonical outcomes, an empirical predictor combines successful policy history and successful leave-
 322 one-out peers. If K keys have total count N , unit pseudocounts and one reserved unseen bucket

323 give

$$\hat{p}(a) = \frac{n(a) + 1}{N + K + 1}, \quad \hat{p}(\text{UNSEEN}) = \frac{1}{N + K + 1}. \quad (5)$$

324 The added detached advantage on an eligible successful row is

$$A_{i,t}^{\text{sem}} = \eta \frac{\min\{-\log \hat{p}(a_i), C\} - \mathbb{E}_{\hat{p}}[\min\{-\log \hat{p}(a), C\}]}{C}, \quad \eta = .10, C = 5. \quad (6)$$

325 Invalid and unsuccessful rows receive zero. The centered score is bounded by η in magnitude. It
 326 can favor an underrepresented verified outcome present in a fresh rollout, but supplies no direct
 327 reinforcement to an absent remembered exemplar. Its heterogeneous empirical effects therefore
 328 remain a comparison with exploration rather than a retention guarantee.

329 For the true conditional distribution $q_\theta = p_\theta(\cdot | C)$,

$$\nabla_\theta H(q_\theta) = \mathbb{E}_{a \sim q_\theta} [(-\log q_\theta(a)) \nabla_\theta \log q_\theta(a)]. \quad (7)$$

330 This motivates the direction of redistribution, not an unbiased-gradient claim for the empirical
 331 estimator. Hold a detached predictor score vector s and baseline b fixed independently of the scored
 332 action, for example conditional on its leave-one-out peers. Averaging successful rows under the
 333 unconditional policy then gives

$$\mathbb{E}_{A \sim p}[\mathbf{1}\{A \in C\}(s_A - b) \nabla_z \log p_A] = P \mathbb{E}_{c \sim q}[s_c \nabla_z \log q_c] + (\mathbb{E}_q[s] - b) \nabla_z P.$$

334 A historical baseline need not equal the current conditional mean, so this bonus can change correctness
 335 as well as diversity. Even with the exact unclipped score $s_c = -\log q_c$ and baseline $b = H(q)$,
 336 averaging over unconditional rollout rows gives $P \nabla H(q)$. For a single correct key, conditional
 337 entropy is zero, yet the implemented unseen bucket can give that key negative semantic advantage.
 338 The comparator thus carries neither an exact entropy-gradient interpretation nor an individual-mode
 339 preservation guarantee.

340 E Empirical Results and Comparisons

341 E.1 Comparison registry

342 The primary comparisons cross fresh objective (Dr.GRPO or MaxRL) with verified replay, first
 343 across three model scales on Level 1 and then at Qwen2.5-0.5B on Level 2. Table 4 separates these
 344 comparisons from the supporting controls and replay-weighting ablation. Terminal populations
 345 use the September 11 frozen census. The registered Qwen2.5-0.5B weighting aggregate retains its
 346 original frozen population; its endpoint values are unchanged in that census.

Table 4: **Comparison scope and available terminal evidence.** A complete block has five registered paired seeds. Primary partial intersections retain their exact counts and receive no interval. UCPO and sparse RLEP-Dr are reported only at the two smaller model scales; their denominators cover all five benchmark domains and five registered seeds at each selected scale.

Comparison	Scope	Terminal coverage	Question
Dr.GRPO / ReplayDr.GRPO	Level 1, all three scales	74/75 pairs; 14/15 complete blocks	Verified memory beyond fresh reward
MaxRL / ReplayMaxRL	Level 1, all three scales	67/75 pairs; 12/15 complete blocks	Memory beyond a stronger fresh objective
Four-method factorial	Level 2, Qwen-0.5B	4/5 complete domains; two additional Pantry Dr.GRPO pairs	Transfer to harder matched tasks
GRPO	Level 1, all three scales	75/75 runs	Reward-standardization control
UCPO	Level 1, 0.5B and 1B	50/50 runs	On-policy advantage redistribution
Sparse RLEP-Dr	Level 1, 0.5B and 1B	47/50 runs	Success memory without uniform key balancing
Fixed Semantic MaxEnt	Level 1, registered scale/domain blocks	10/15 complete blocks	Fresh-rollout semantic exploration
Uniform / frequency replay	Level 1, nine registered blocks	43/45 treatment cells; 7/9 complete blocks	Contribution of replay-key balancing

E.2 Terminal comparisons across scales

Within each model–domain–seed comparison, methods share initialization, data identities, the eight-pass horizon, evaluation requests, verifier, and replay scheduling code. The replay control runs the same bank and scoring path with zero replay derivative. Resulting policies can build different banks. The admissible intersections are defined in Appendix L.1.

Table 5: Matched results after eight training passes. Columns compare Dr.GRPO with ReplayDr.GRPO after exactly eight training passes. Entries are means over the displayed paired-seed intersection; $n@pass$ gives its size and training pass. P, M, and D denote $pass@8$, $mean@8$, and $distinct@8$. Every available source-admissible terminal pair is included, no shallower checkpoint is carried forward, and no missing result is imputed. Falcon Countdown uses four admissible seeds (Appendix L.1). Pale green marks a ReplayDr.GRPO mean that is strictly higher than its matched Dr.GRPO mean. Archived models: Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B.

Model	Domain	$n@pass$	Dr.GRPO			ReplayDr.GRPO		
			P	M	D	P	M	D
Qwen2.5-0.5B	Graph coloring	5@8.0	.323	.323	.325	.969	.404	2.438
	Countdown	5@8.0	.480	.465	.487	.672	.597	1.637
	Python factors	5@8.0	.172	.172	.172	.528	.507	.571
	MathIR	5@8.0	.512	.464	.513	.796	.688	.829
	PantryPlan	5@8.0	.522	.522	.522	.729	.545	1.539
Falcon3-1B	Graph coloring	5@8.0	.656	.300	1.034	.809	.334	1.717
	Countdown	4@8.0	.325	.267	.415	.632	.343	.838
	Python factors	5@8.0	.000	.000	.000	.979	.722	1.107
	MathIR	5@8.0	.095	.068	.097	.736	.292	.756
	PantryPlan	5@8.0	.633	.633	.633	.751	.560	1.576
Qwen2.5-3B	Graph coloring	5@8.0	.682	.360	1.060	.836	.349	1.637
	Countdown	5@8.0	.609	.569	.628	.746	.505	.962
	Python factors	5@8.0	.069	.069	.069	.659	.603	.852
	MathIR	5@8.0	.278	.168	.279	.686	.580	.706
	PantryPlan	5@8.0	.638	.558	.841	.744	.546	1.514

Figure 8 resolves the domain-specific results behind the main cross-domain display. The 3B MaxRL pairs use Graph/Countdown/Python/MathIR/Pantry counts 5/3/5/3/1; the Dr.GRPO pair uses five seeds except Falcon Countdown, which uses four. These populations remain separate: a factorial interaction requires the common four-arm intersection, not differences of aggregates with different seeds.

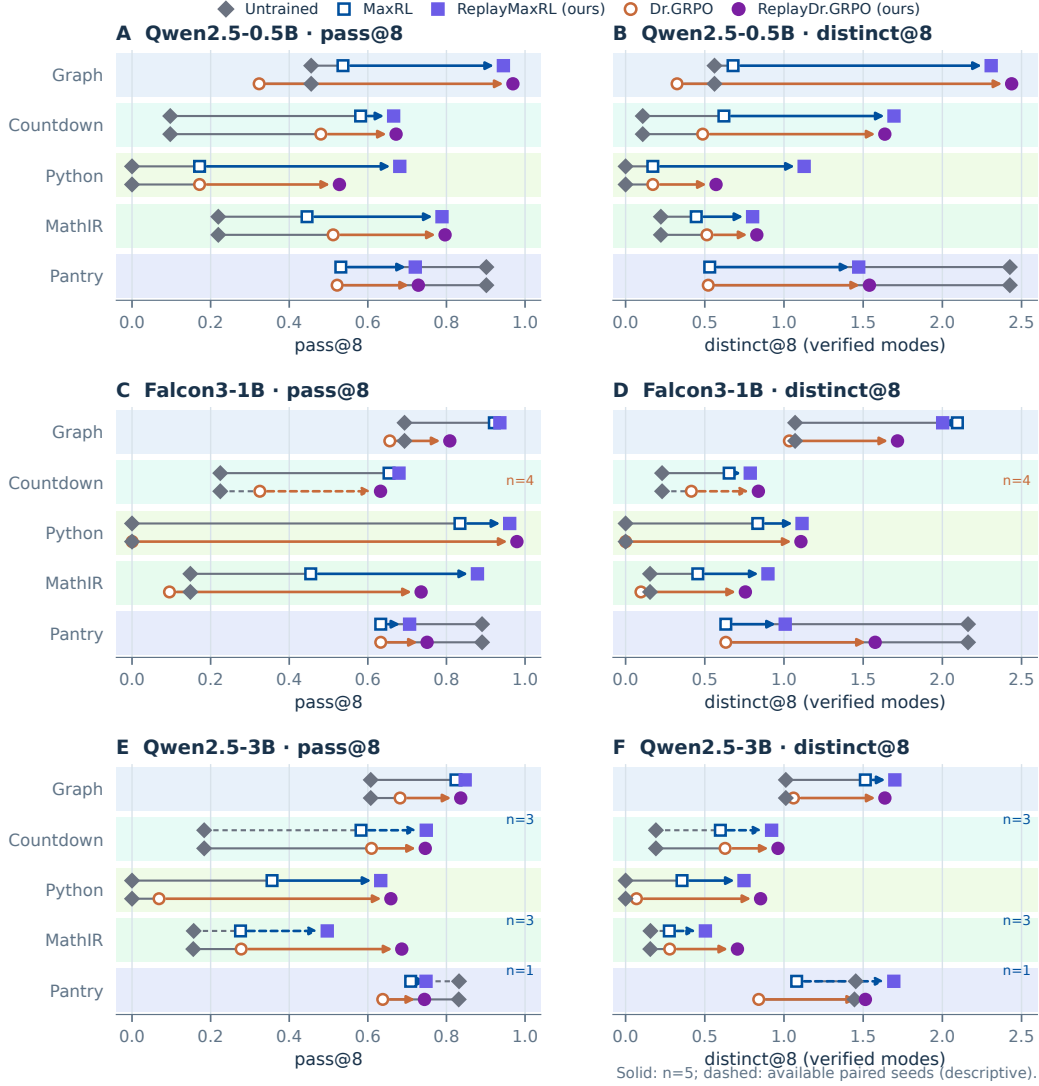


Figure 8: **Per-domain results behind the cross-domain averages in Figure 5.** Rows show Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns show correctness and raw support. MaxRL–ReplayMaxRL pairs retain five terminal seeds throughout the two smaller models. Qwen2.5-3B uses $n = 5, 3, 5, 3, 1$ in domain order; dashed connectors mark partial blocks. Dr.GRPO–ReplayDr.GRPO uses four admissible seeds for Falcon Countdown and five elsewhere. Domain tints match Figure 2. Replay arrows, trained-arm markers, and untrained references use the same encoding as the main figure. The E118 model index groups the archived exports by model and domain.

Write $P = \text{pass}@8$, $D = \text{distinct}@8$, and $B = D - P$. The derived B counts verified modes beyond the first correct outcome in each sampled set and remains coupled to correctness. The endpoint tables report exact paired terminal effects. A dash denotes no observed pair; partial means retain $n/5$ without an interval. Complete-block intervals are nominal 95% estimates, without multiple-comparison or repeated-look adjustment. All underlying arm means and seed effects remain in the adjacent machine-readable records.

Table 6: **ReplayMaxRL effects across model scales at pass 8.** All effects are ReplayMaxRL minus MaxRL on the displayed paired seeds. Brackets give paired Student- t intervals for ΔB in complete blocks. The machine-readable report also retains both arm means, all per-seed effects, and intervals for P and D .

Model	Domain	Replay arm	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	ReplayMaxRL	5/5	+0.408	+1.630	+1.221 [+0.926, +1.517]
Qwen-0.5B	Countdown	ReplayMaxRL	5/5	+0.084	+1.074	+0.990 [+0.859, +1.121]
Qwen-0.5B	Python	ReplayMaxRL	5/5	+0.509	+0.955	+0.446 [-0.156, +1.048]
Qwen-0.5B	MathIR	ReplayMaxRL	5/5	+0.343	+0.356	+0.013 [+0.008, +0.017]
Qwen-0.5B	PantryPlan	ReplayMaxRL	5/5	+0.189	+0.941	+0.751 [+0.520, +0.982]
Falcon-1B	Graph	ReplayMaxRL	5/5	+0.014	-0.093	-0.107 [-0.220, +0.005]
Falcon-1B	Countdown	ReplayMaxRL	5/5	+0.025	+0.133	+0.108 [+0.082, +0.134]
Falcon-1B	Python	ReplayMaxRL	5/5	+0.127	+0.280	+0.153 [-0.272, +0.578]
Falcon-1B	MathIR	ReplayMaxRL	5/5	+0.424	+0.443	+0.019 [+0.006, +0.031]
Falcon-1B	PantryPlan	ReplayMaxRL	5/5	+0.073	+0.375	+0.302 [+0.034, +0.569]
Qwen-3B	Graph	ReplayMaxRL	5/5	+0.023	+0.186	+0.163 [+0.021, +0.305]
Qwen-3B	Countdown	ReplayMaxRL	3/5	+0.166	+0.324	+0.158
Qwen-3B	Python	ReplayMaxRL	5/5	+0.277	+0.391	+0.115 [+0.080, +0.149]
Qwen-3B	MathIR	ReplayMaxRL	3/5	+0.221	+0.228	+0.007
Qwen-3B	PantryPlan	ReplayMaxRL	1/5	+0.039	+0.613	+0.574

At Qwen2.5-0.5B, ReplayMaxRL adds 1.221, .990, and .751 extra modes on Graph, Countdown, and PantryPlan, respectively; each five-seed interval excludes zero. Falcon Graph instead has a negative mean raw-support effect (-.093, interval [-.229, +.043]), and its correctness effect is inconclusive (+.014, [-.016, +.044]). Positive cross-domain effects therefore do not imply a breadth gain in every domain. The complete Qwen2.5-3B Graph and Python blocks have positive extra-mode intervals while their correctness intervals include zero. The three partial 3B domains retain their own seed sets; the main figure’s equally weighted domain average is descriptive.

E.3 Terminal comparison on the harder level

Figure 6 uses Graph, Countdown, Python, and MathIR: each has all five seeds at step 3,072 for both levels and all four methods. Means average those four domains equally within each seed, then the five seed means. This is a descriptive 20-cell comparison, not a five-domain estimate; the levels have separate held-out prompts, with the population differences in Appendix B.3. At Level 2, PantryPlan has three Dr.GRPO, three ReplayDr.GRPO, one MaxRL, and one ReplayMaxRL terminal endpoint. Its only terminal pairs are Dr.GRPO/replay seeds 43 and 46, with no common four-arm seed. Independently available arm means do not estimate paired effects.

Table 7: **Replay effects on the harder Level-2 tasks at pass 8.** MaxRL and Dr.GRPO identify the fresh objective; each effect is its replay variant minus the corresponding objective without replay. Each row uses its own pairwise intersection, which can exceed the four-method intersection. Brackets give paired Student-*t* intervals for ΔB only when all five registered pairs are available.

Model	Domain	Objective	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	Dr.GRPO	5/5	+0.561	+1.019	+0.458 [+0.394, +0.522]
Qwen-0.5B	Graph	MaxRL	5/5	+0.305	+0.668	+0.363 [+0.280, +0.446]
Qwen-0.5B	Countdown	Dr.GRPO	5/5	+0.025	+0.045	+0.020 [-0.014, +0.055]
Qwen-0.5B	Countdown	MaxRL	5/5	+0.040	+0.057	+0.017 [+0.001, +0.032]
Qwen-0.5B	Python	Dr.GRPO	5/5	+0.075	+0.400	+0.325 [-0.228, +0.878]
Qwen-0.5B	Python	MaxRL	5/5	+0.037	+0.212	+0.175 [-0.295, +0.645]
Qwen-0.5B	MathIR	Dr.GRPO	5/5	+0.493	+0.493	+0.000 [+0.000, +0.000]
Qwen-0.5B	MathIR	MaxRL	5/5	+0.173	+0.175	+0.002 [-0.003, +0.006]
Qwen-0.5B	PantryPlan	Dr.GRPO	2/5	+0.234	+0.367	+0.133
Qwen-0.5B	PantryPlan	MaxRL	0/5	—	—	—

Level-2 Graph provides the clearest replay effects on correctness and extra modes under both objectives. Countdown’s ReplayMaxRL extra-mode effect is small (+.017, interval [+0.001, +0.032]), while its correctness and raw-support intervals include zero. Python and MathIR have positive mean correctness and raw-support effects with intervals including zero; MathIR’s extra-mode effects are near zero. Pantry’s two-seed Dr.GRPO replay contrast is +.234 on *P*, +.367 on *D*, and +.133 on *B*, without an interval.

For each common seed and endpoint *Y*, the replay interaction is

$$I = (Y^{\text{ReplayMaxRL}} - Y^{\text{MaxRL}}) - (Y^{\text{ReplayDr.GRPO}} - Y^{\text{Dr.GRPO}}).$$

Replay can add value under both fresh objectives without a positive interaction. On Level-2 Graph, MaxRL alone raises *P* relative to Dr.GRPO by +.233 ([+.140, +.325]), while the replay interaction on *P* is −.256 ([-.329, −.183]). This is an additional replay benefit after stronger fresh optimization, without a superadditive interaction.

Table 8: **Fresh-objective effects and replay interactions on Level 2.** All three contrasts within a domain use the same four-arm seed intersection. The interaction is (ReplayMaxRL − MaxRL) − (ReplayDr.GRPO − Dr.GRPO). Each entry gives its mean and, only for a complete five-seed block, a paired unadjusted 95% Student-*t* interval. Archived four-arm exports: E119 model index.

Domain	n	Contrast	ΔP	ΔD	ΔB
Graph	5	MaxRL − Dr.GRPO	+.233 [+0.140, +0.325]	+.322 [+0.162, +0.482]	+.089 [+0.019, +0.159]
Graph	5	ReplayMaxRL − ReplayDr.GRPO	−.023 [−0.062, +0.016]	−.029 [−0.198, +0.140]	−.006 [−0.142, +0.130]
Graph	5	Replay × MaxRL interaction	−.256 [−0.329, −0.183]	−.351 [−0.446, −0.255]	−.095 [−0.186, −0.003]
Countdown	5	MaxRL − Dr.GRPO	−.014 [−0.073, +0.045]	−.014 [−0.073, +0.045]	+0.000 [+0.000, +0.000]
Countdown	5	ReplayMaxRL − ReplayDr.GRPO	+0.002 [−0.008, +0.012]	−.002 [−0.047, +0.044]	−.004 [−0.042, +0.035]
Countdown	5	Replay × MaxRL interaction	+0.016 [−0.047, +0.078]	+0.012 [−0.068, +0.092]	−.004 [−0.042, +0.035]
Python	5	MaxRL − Dr.GRPO	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]
Python	5	ReplayMaxRL − ReplayDr.GRPO	−.037 [−0.232, +0.157]	−.188 [−0.988, +0.613]	−.150 [−1.014, +0.713]
Python	5	Replay × MaxRL interaction	−.037 [−0.232, +0.157]	−.188 [−0.988, +0.613]	−.150 [−1.014, +0.713]
MathIR	5	MaxRL − Dr.GRPO	+.283 [−0.193, +0.759]	+.283 [−0.193, +0.759]	+0.000 [+0.000, +0.000]
MathIR	5	ReplayMaxRL − ReplayDr.GRPO	−.036 [−0.158, +0.085]	−.035 [−0.158, +0.088]	+0.002 [−0.003, +0.006]
MathIR	5	Replay × MaxRL interaction	−.320 [−0.864, +0.225]	−.318 [−0.860, +0.224]	+0.002 [−0.003, +0.006]
PantryPlan	0	MaxRL − Dr.GRPO	—	—	—
PantryPlan	0	ReplayMaxRL − ReplayDr.GRPO	—	—	—
PantryPlan	0	Replay × MaxRL interaction	—	—	—

E.4 Accuracy and verified modes throughout training

The following curves include every primary method and model scale used in the main paper. Each objective pair uses its domain-specific terminal seed intersection as one fixed cohort throughout

392 training. A checkpoint is plotted only when all members have complete, valid evaluations for both
 393 methods. Missing or conflicting checkpoints leave gaps; seed counts never shrink to extend a mean.
 394 Four fixed-seed draws of eight samples on the same 128 held-out prompts define each evaluation.
 395 Accuracy and mode curves use identical seeds and checkpoints, without smoothing or outcome-based
 396 selection.

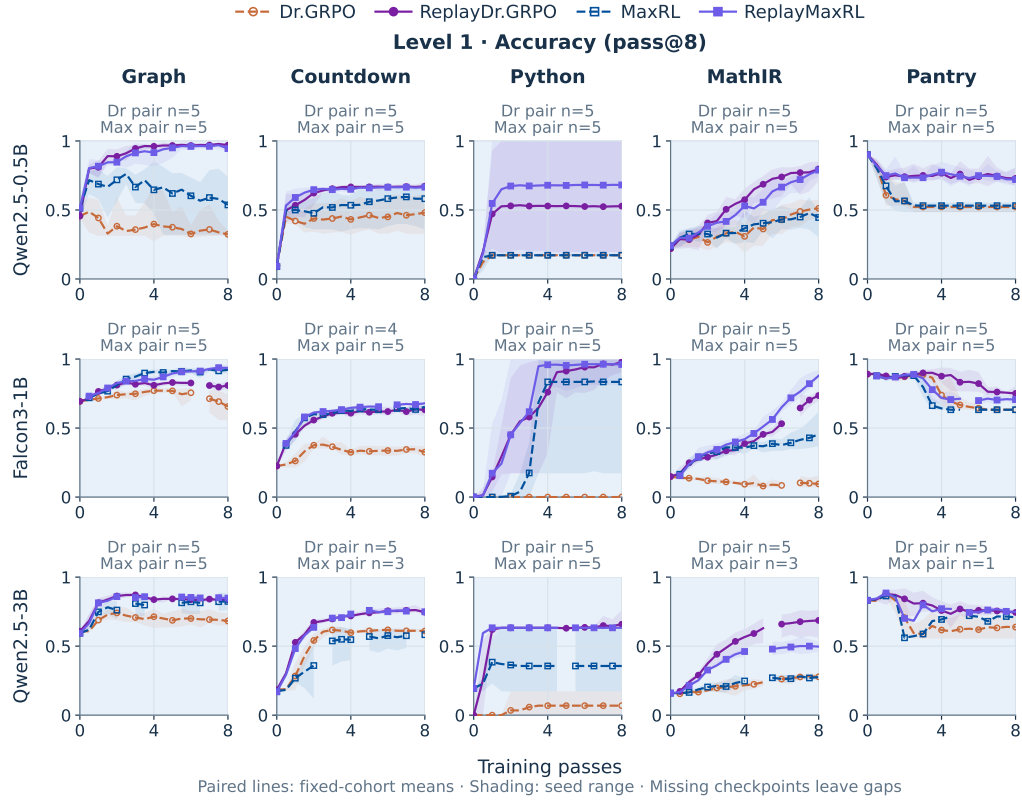


Figure 9: **Accuracy throughout training at all three model scales.** Rows are Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns are the five domains. Dr.GRPO, ReplayDr.GRPO, MaxRL, and ReplayMaxRL use the same method colors as Figure 5. Lines are fixed-cohort means of pass@8; bands are seed ranges, not confidence intervals. Counts identify each objective’s terminal paired seeds. Qwen2.5-3B MaxRL retains Graph/Countdown/Python/MathIR/Pantry counts 5/3/5/3/1; Falcon Countdown Dr.GRPO/replay uses four. Gaps indicate missing or conflicting checkpoints.

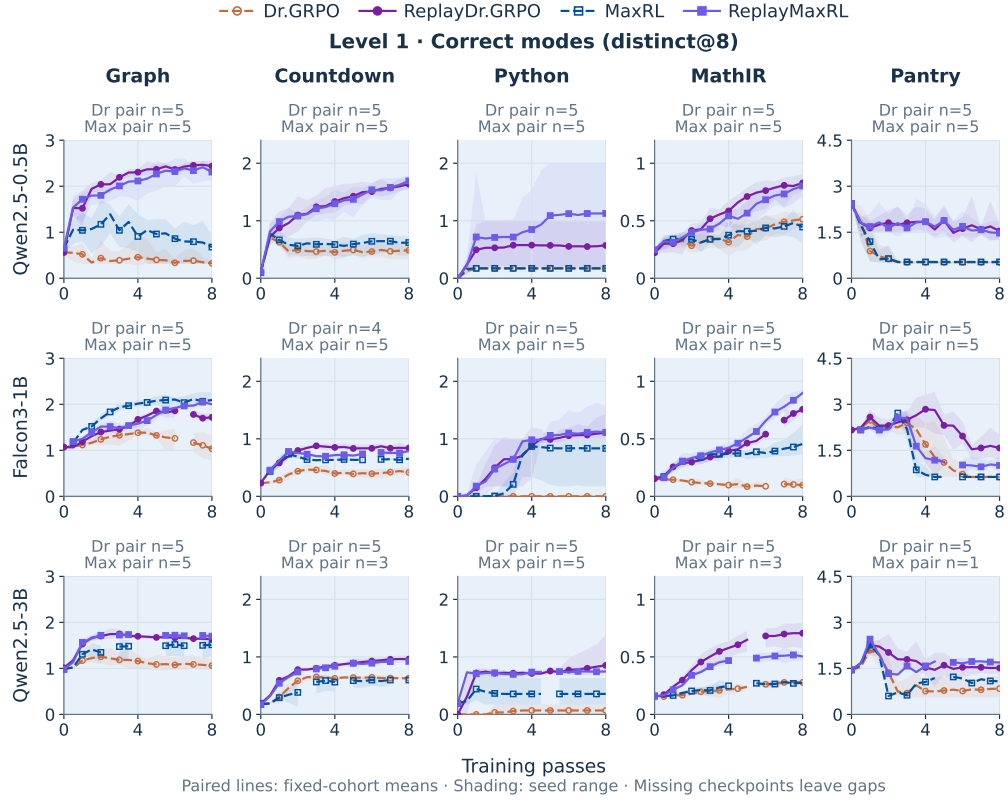


Figure 10: **Verified modes throughout training at all three model scales.** The same model rows, methods, seeds, and checkpoints as Figure 9, now measuring distinct@8. Lines are descriptive fixed-cohort means and bands show seed ranges. A count of one has no band. Domain-specific vertical scales are shared across model rows; absolute heights across different domains should be read against their labeled axes. Missing evaluations are never carried forward or joined across a gap.

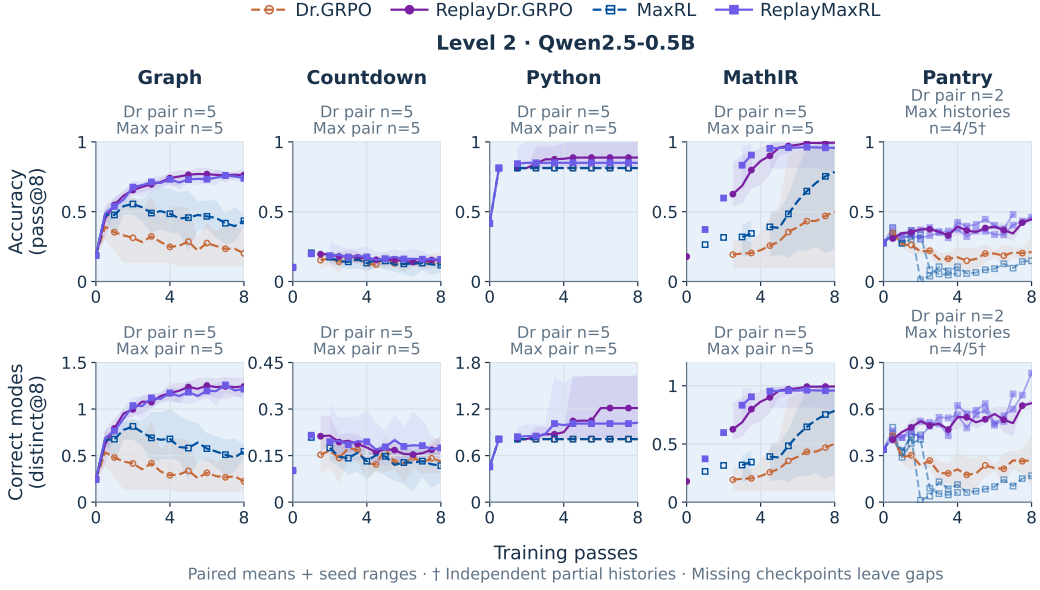


Figure 11: **Accuracy and verified modes during Level-2 training.** Qwen2.5-0.5B is the model tested at this level. Rows show pass@8 and distinct@8 for all four primary methods. Graph, Countdown, Python, and MathIR contain five terminal paired seeds per objective. PantryPlan’s Dr.GRPO/replay means use their two terminal paired seeds; additional MaxRL and ReplayMaxRL histories are individual, unpaired observations, marked separately without a mean or band. They do not estimate a paired replay effect. Both metrics follow the same evaluation rules, with gaps at missing checkpoints and no extrapolation to later training passes.

397 E.5 Supporting comparators

398 GRPO changes group-reward normalization; UCPO redistributes fresh on-policy advantage; sparse
 399 RLEP-Dr replays verified successes without uniform canonical-key weighting. Fixed Semantic
 400 MaxEnt supplies a fresh-rollout exploration incentive (Appendix D.5). These are controls for
 401 alternative explanations of the retention results.

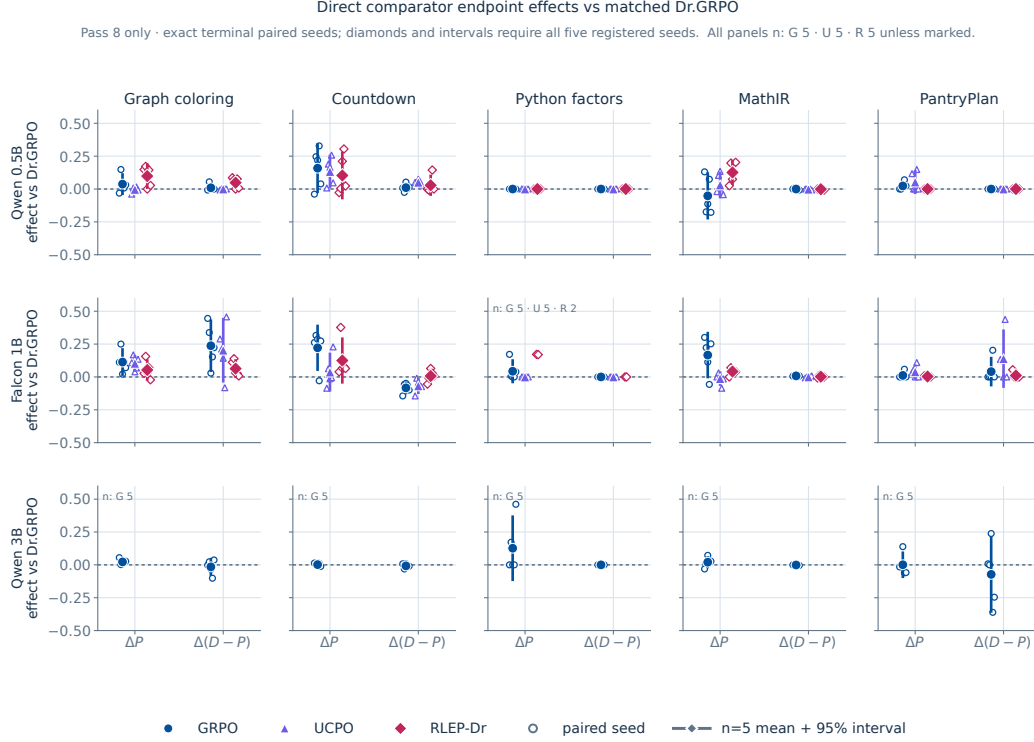


Figure 12: **Direct alternatives produce local rather than uniform final gains.** Each effect subtracts the same-seed Dr.GRPO control from the comparator after eight passes. Effects are shown for pass@8 and extra verified modes beyond the first success. Circles are paired seeds; diamonds and 95% Student- t intervals appear only for complete five-seed blocks. Exact n is printed, and unsupported comparisons remain blank rather than being extrapolated or imputed at any seed.

402 The alternatives give local rather than uniform benefits. Qwen-0.5B Countdown UCPO raises
 403 P by $+0.133$ ($[+0.005, +0.262]$) and B by $+0.051$ ($[+0.031, +0.071]$), whereas its MathIR effects are
 404 inconclusive. MathIR RLEP-Dr raises P by $+0.126$ ($[+0.030, +0.222]$) with neutral extra modes
 405 ($-0.002, [-0.006, +0.003]$); its Countdown effects and Falcon Pantry effects are inconclusive. The
 406 full five-seed 3B plain-GRPO blocks likewise have intervals including zero on both correctness and
 407 extra modes. RLEP's frozen offline pool was collected at temperature $.7$, top- $p = .95$; all reported
 408 evaluations use the common temperature-1, top- $p = 1$ contract. An offline pool versus an online
 409 bank remains an algorithmic difference.

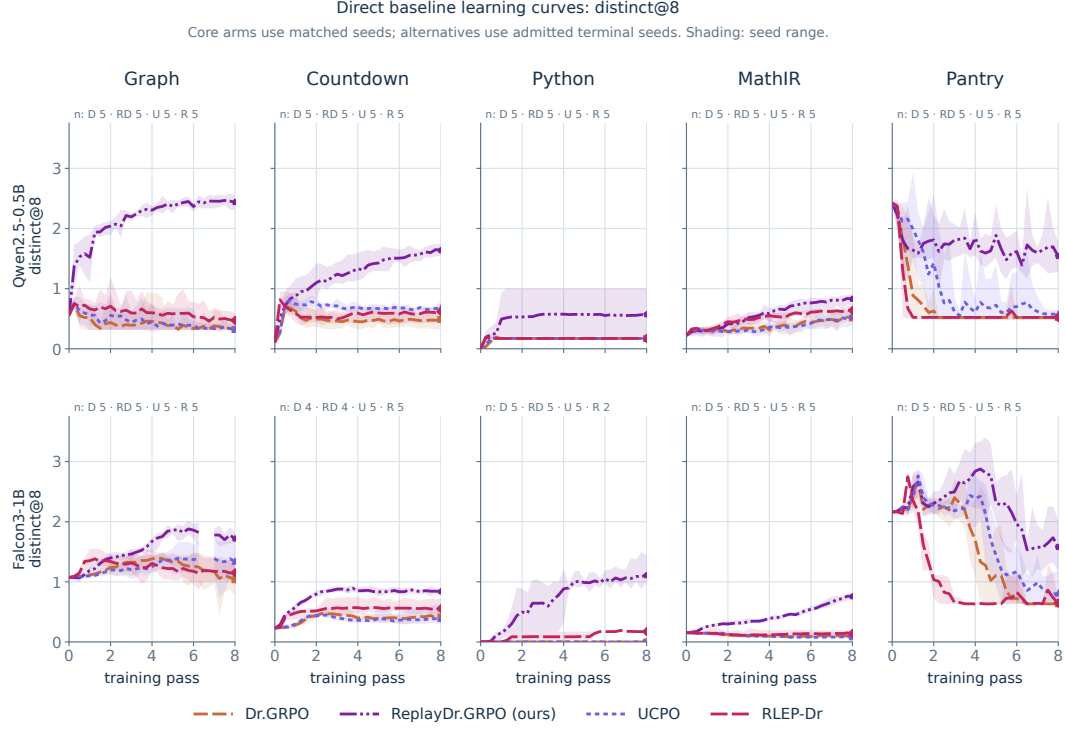


Figure 13: **Verified-mode trajectories for the supporting comparators.** Rows show Qwen2.5-0.5B and Falcon3-1B; columns are the five domains. UCPO and sparse RLEP-Dr are compared with Dr.GRPO and ReplayDr.GRPO at these two smaller scales. Labels give exact method-specific seed counts. Lines are descriptive distinct@8 means and bands are seed ranges. The corresponding accuracy curves use identical populations in Figure 14.

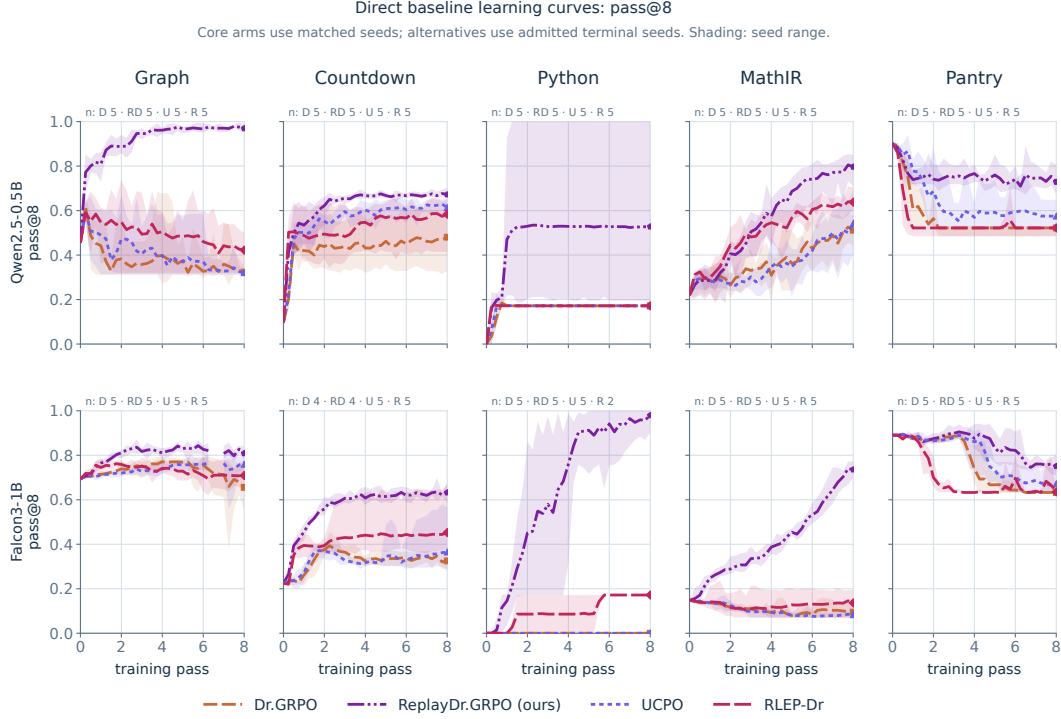


Figure 14: **Accuracy trajectories for the supporting comparators.** The same model rows, domains, methods, terminal seeds, and registered checkpoints as Figure 13, now measuring pass@8. Bands show the seed range and the vertical axes span $[0, 1]$ throughout. MaxRL and ReplayMaxRL at all three scales are compared on both metrics in Figures 9 and 10.

410 E.6 Uniform versus fresh-frequency replay

411 The weighting ablation retains the same bank construction and update code, changing only the within-
 412 bank target: uniform weights over retained keys versus weights proportional to validator-positive
 413 fresh-rollout counts. Policy-dependent bank contents can subsequently differ. Positive contrasts below
 414 favor uniform weighting. The registered co-primary contrasts are ΔB and ΔP ; the five-domain
 415 Qwen2.5-0.5B aggregate first averages domain effects within each matched seed, giving equal domain
 416 weight.

417 Complete-block intervals enumerate all $5^5 = 3,125$ ordered bootstrap resamples of the five paired
 418 training seeds and use linearly interpolated 2.5th and 97.5th percentiles. A resampled seed carries its
 419 complete domain vector when estimating the five-domain aggregate. The endpoints and aggregate
 420 were preregistered; the exact seed-resampling and percentile conventions are analysis choices. These
 421 nominal intervals quantify seed variability within the fixed benchmark, without resampling prompts
 422 or domains or adjusting across comparisons.

Table 9: **Uniform versus fresh-frequency replay at pass 8.** Effects are uniform minus frequency weighting. Brackets give the paired percentile bootstrap interval for ΔB over all 5^5 ordered resamples in complete blocks. Every observed treatment endpoint in this snapshot passes the persisted replay-weight telemetry audit. Partial blocks retain exact seed counts. Paired model exports.

Model	Domain	Weighting	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	Uniform	5/5	+0.066	+0.662	+0.596 [+0.513, +0.687]
Qwen-0.5B	Countdown	Uniform	5/5	+0.022	+0.661	+0.639 [+0.544, +0.733]
Qwen-0.5B	Python	Uniform	5/5	-0.077	-0.040	+0.037 [-0.012, +0.123]
Qwen-0.5B	MathIR	Uniform	5/5	-0.005	+0.014	+0.019 [-0.001, +0.038]
Qwen-0.5B	PantryPlan	Uniform	5/5	+0.043	+0.341	+0.298 [+0.174, +0.400]
Falcon-1B	Graph	Uniform	5/5	+0.045	+0.278	+0.233 [+0.182, +0.320]
Falcon-1B	PantryPlan	Uniform	5/5	-0.021	-0.167	-0.146 [-0.564, +0.414]
Qwen-3B	Graph	Uniform	4/5	-0.006	+0.063	+0.070
Qwen-3B	PantryPlan	Uniform	4/5	+0.011	+0.232	+0.222

The registered Qwen2.5-0.5B five-domain effect is +.318 extra modes per prompt ([+.272, +.358]), with positive within-seed domain averages in all five seeds. Graph, Countdown, and PantryPlan drive this result; Python and MathIR remain inconclusive. The correctness effect is +.010 ([-.119, +.139]), establishing neither a gain nor equivalence. Python’s correctness effects range from -.781 to +.828 across seeds despite small extra-mode effects in four seeds.

Falcon Graph also favors uniform weighting on extra modes (+.233, [+0.182, +0.320]), but its mean sampled correctness decreases by -.0152 ([-.0214, -.0090]); its P interval includes zero. Falcon Pantry is inconclusive: $\Delta B = -.146$ ([-.564, +.414]) and $\Delta P = -.021$ ([-.063, +.026]). Qwen2.5-3B Graph and Pantry have descriptive extra-mode effects of +.070 and +.222 at four pairs each. All observed treatment endpoints pass the persisted replay-weight telemetry audit. The evidence supports a contribution from key balancing in the stated domains, not a uniform accuracy or breadth advantage across every model–domain block.

Evidence boundary. The primary effects are within-domain, within-scale paired comparisons. They do not establish a scaling law, a positive factorial interaction, or preservation of every previously discovered mode. Partial domains are reported as observed and do not support complete-block generality.

F Collapse and Replay Mechanism Checks

F.1 Breadth before and after verifier-only training

The problem precheck evaluates unmodified Dr.GRPO and GRPO at passes 0 and 8 on all three models and five domains: 150 complete runs, each with four fixed $K = 8$ draws. The five-domain macro-average of extra verified modes falls by 98.2%–99.4% at Qwen2.5-0.5B and 41.5%–71.6% at larger scales. Correctness falls in 10 of the 30 domain–method–scale comparisons, while macro raw `distinct@8` ends above its initial value at Qwen2.5-3B. This distinguishes loss of multiple correct outcomes from the broader question of whether any correct outcome is sampled.

Initial evaluations are measured within each method. Qwen2.5-3B MathIR and Pantry E80-R1 seeds 71–74 carry slightly different initial requests from plain GRPO (.154 versus .162, and .832 versus .828 on P). Those cells therefore have separate initial references.

Graph and PantryPlan contain substantial initial breadth at every scale; the other three domains begin with at most .011 extra modes per prompt. A retained-breadth percentage is therefore reported only when the initial extra-mode count exceeds .05. Raw `distinct@8` ends below its initial value in six of the 15 Dr.GRPO domains. ReplayDr.GRPO does so in two available domain summaries, both PantryPlan; Falcon Countdown uses four admissible seeds. These observations concern sampled held-out support, not the survival histories of particular stored modes.

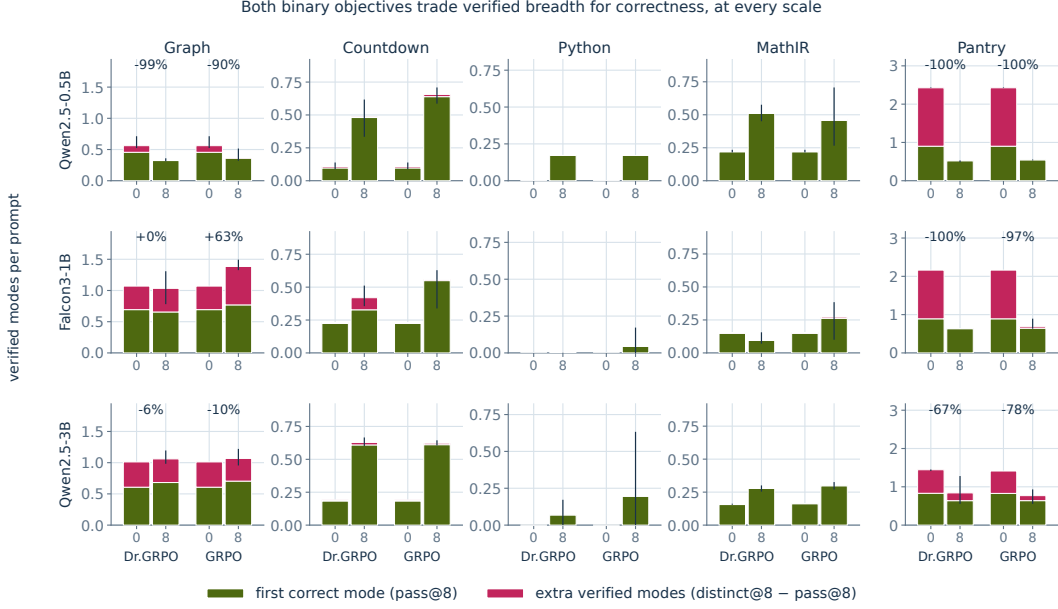


Figure 15: **Verifier-only training reduces average extra verified modes at each tested scale.** Bars split distinct@8 into the first correct mode and every verified mode beyond it, at training pass 0 and pass 8, for unmodified Dr.GRPO then GRPO in each panel. All 150 evaluations are complete at five seeds and four fixed $K = 8$ draws; whiskers are the seed range and pass 0 is measured per method. Badges give the change in extra verified modes, printed only where pass-0 breadth exceeds .05 of them; the other three domains start at .011 or less.

Table 10: **Problem precheck, five-domain macro-averages.** Both verifier-only objectives at pass 0 and pass 8, all three scales, five seeds and four fixed $K = 8$ draws per model–domain–method comparison. “Extra modes” is distinct@8–pass@8; “removed” is the fraction of that quantity lost between pass 0 and pass 8. Breadth loss is present at every scale and is far from uniform across scales, and at Falcon3-1B Dr.GRPO macro pass@8 falls as well.

Model	Dr.GRPO			GRPO		
	pass@8	extra modes	removed	pass@8	extra modes	removed
Qwen2.5-0.5B	0.335 → 0.402	0.329 → 0.002	99.4%	0.335 → 0.435	0.329 → 0.006	98.2%
Falcon3-1B	0.391 → 0.342	0.332 → 0.094	71.6%	0.391 → 0.454	0.332 → 0.135	59.4%
Qwen2.5-3B	0.356 → 0.455	0.205 → 0.120	41.5%	0.356 → 0.490	0.199 → 0.101	49.4%

Uniform-action references are meaningful for the enumerable Graph and Pantry representations. They are not defined for open-ended programs or derivations. Pantry’s bit-uniform reference exceeds every final model, so this domain is a support-concentration stress test rather than evidence of complete support coverage.

F.2 Fresh-gradient degeneracy and observed output collapse

A Dr.GRPO group with identical rewards has exactly zero fresh task gradient. Only mixed groups containing both correct and incorrect responses can supply fresh reward information; optimizer momentum can still move a policy when that task gradient is zero. The fractions of mixed groups over all 3,072 Qwen2.5-0.5B updates, averaged over five seeds, are:

Table 11: **Absolute references for raw distinct@8.** *Uniform* samples the exposed action representation uniformly and is defined only for Graph (27 color strings) and PantryPlan (64 bit masks). Frozen is the matched pass-0 model; the final two columns are pass-8 means. Dashes avoid pretending that an open-ended program or derivation space has a canonical uniform distribution.

Model	Domain	Uniform	Frozen	Dr.GRPO	ReplayDr.GRPO
Qwen2.5-0.5B	Graph coloring	1.633	.561	.325	2.438
	PantryPlan	2.153	2.426	.522	1.539
Falcon3-1B	Graph coloring	1.633	1.070	1.034	1.717
	PantryPlan	2.153	2.162	.633	1.576
Qwen2.5-3B	Graph coloring	1.633	1.012	1.060	1.637
	PantryPlan	2.153	1.445	.841	1.514

Domain	Dr.GRPO	ReplayDr.GRPO
Graph coloring	.144	.954
Countdown	.140	.456
Python factors	.015	.185
MathIR	.364	.453
PantryPlan	.123	.789

Thus 98.5% of Python control updates carry zero fresh task gradient. Replay supplies a verified-likelihood gradient directly and can also change the policy so that later fresh groups become mixed.

A raw-response audit of all five Qwen2.5-0.5B Python control seeds establishes observed output collapse directly. At pass 0, a draw has 7.99 unique raw completions per prompt on average. By steps 288–480, every seed produces 32 byte-identical temperature-1 samples per prompt, also identical to the logged greedy completion, on all 128 prompts. This persists through step 3,072. Terminal draws contain 1.00 unique completion per prompt in the control and 1.97 under replay. A collapsed completion can be incorrect: this is one observed completion per prompt, not one correct key everywhere. Aggregate endpoint equality or zero draw spread alone would not establish that conclusion, nor does this finite audit rule out low-probability tails.

These checks explain a vulnerability of the registered sparse binary-reward recipe. They do not establish dominance over methods that keep fresh on-policy signal active. Dynamic mixed-group sampling, larger groups, reference KL, token-entropy regularization, and wider-decoding evaluations remain relevant controls beyond the registered factorial.

F.3 Fixed-bank exemplar score trajectories

The fixed-bank study tracks Qwen2.5-0.5B ReplayDr.GRPO on Graph for seeds 43–47. Before step 384’s fresh group, bank membership and fresh counts are frozen; admission then stops while uniform replay continues through 3,072 updates under the same training recipe. Every frozen exemplar is identified by prompt and canonical-key fingerprints, without selection by its score trajectory. Each scheduled replay visit records mean token log probability s and sequence log probability ℓ . Changes compare an identity’s first and final post-freeze visits; observation steps can differ across identities.

All 3,406 frozen identities across 1,575 seed–prompt pairs have finite, aligned scores at every scheduled visit: 8–9 visits per identity and 29,041 observations in total. The pooled median change is $+0.038$ nat/token (95% interval $[+.025, +.048]$), but the 10th percentile is $-.256$ and 2.61% of identities lose more than $.5$ nat/token ($[1.54, 3.77]\%$). The worst intermediate change is -1.328 nat/token. On the sequence scale, the median is $+0.154$ nat/sequence ($[+.104, +.198]$), while 21.99% lose more than $.5$ nat/sequence. Lengths range from 4 to 100 tokens, so the two threshold percentages have different meanings.

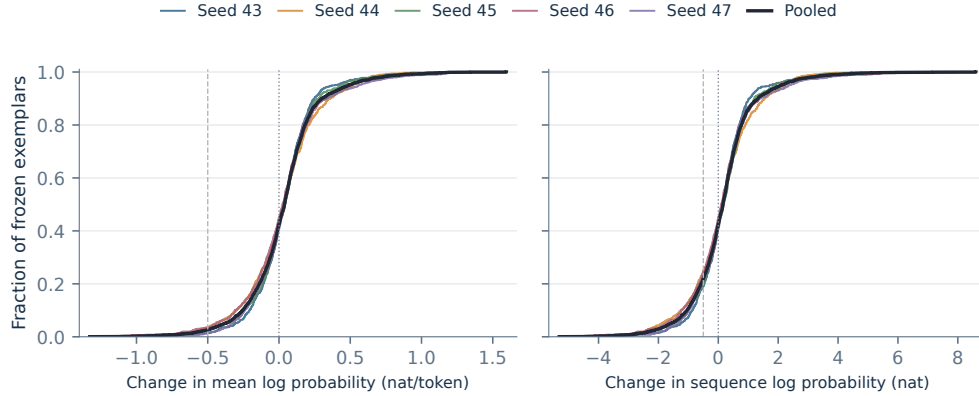


Figure 16: **Fixed-exemplar scores improve centrally but retain a declining tail.** Empirical cumulative distributions show first-to-final post-freeze score changes for every frozen identity, with one curve per seed and the pooled identity distribution. Mean token and sequence scores are separate panels; positive values indicate increased teacher-forced likelihood. Neither panel measures the exact probability of generating a canonical mode during policy sampling.

Pooled summaries weight identities equally. Descriptive 95% intervals use 10,000 hierarchical bootstrap draws (random seed 121), resampling prompts within each original seed and then five seeds with replacement; repeated seeds reuse their within-draw prompt sample. Linear empirical quantiles and percentile intervals are analysis choices. The artifact `paper/results/e121_fixed_bank_survival.json` retains all seed summaries, identity trajectories, source hashes, and audit diagnostics.

These are teacher-forced score surrogates, not exact probabilities of sampling canonical modes. Visits do not cover every intervening update, and the declining tails exclude a claim that every exemplar improves. Without a matched no-replay fixed-bank arm, this study does not identify a causal replay effect. It neither numerically validates Theorem K.6’s probability floor nor certifies a finite-network guarantee. It adds longitudinal measurements of fixed exemplars to the held-out support evidence.

G Inference-Only Concentration in a Hosted Model

This exploratory evaluation measures the current output distribution of GPT-5.6 Sol under the inherited MODEBENCH interface. It is separate from the matched training interventions: it adds no frontier-model training, replay comparison, or controlled model-size contrast. Only the completed, audited deployment enters these tables.

Protocol and provenance. The September 11, 2026 run uses the Azure deployment `gpt-5.6-sol`; saved provider headers identify snapshot `gpt-5.6-sol-2026-07-09`. Each of the five domains contributes 128 held-out prompts at each of three levels. Eight stateless requests per prompt yield 15,360 responses. Requests use medium reasoning, an 8,192-token output limit, no tools and no conversation state. The service returns temperature 1.0 and top- p .98; neither setting was overridden. These settings and the hosted natural-language interface differ from the local training evaluation’s token-level grammar constraints.

Level 1 uses the native E118 evaluation splits retained in E124. Level 2 uses the frozen matched r5 construction; Level 3 uses the matched v3 construction admitted by the completed canonical v3 confirmation. Level 3 is an exploratory test extension here, not a third level of the paper’s training intervention. All cells use the multi-answer split. The Level-2/Level-3 support histograms match a separate fixed-reference Level-1 confirmation reserve, which differs from the native Level-1 split in Graph, Countdown and Python. Prompts differ across levels and are not paired by row index. Calibration against small-model references does not guarantee increasing difficulty for this hosted model.

Exact prompts, dataset identities and hashes, requests, provider-exposed responses, grades, retry receipts, usage and verifier snapshots are retained in the experiment’s evidence directory. All 15,360

intended samples completed, with distinct response identifiers; 107 failed API attempts were retained and recovered by retries. Provider random-number independence and hidden reasoning contents cannot be audited from the exposed responses.

Table 12: **Hosted correctness and concentration under strict grading.** GPT-5.6 Sol, with medium reasoning. Each level weights five domains equally. M and P are mean@8 and pass@8 (%); D is raw distinct@8. C is the percentage of within-prompt correct pairs sharing a mode, pair weighted within each domain. Brackets give pointwise 95% intervals from whole-prompt resampling.

Level	M	P	D	C (95% interval)
1	83.0	95.8	1.71	74.0 [71.9, 76.1]
2	62.9	77.5	1.16	82.3 [80.3, 84.4]
3	68.7	80.3	1.18	84.5 [82.8, 86.2]

Metrics and uncertainty. Per-response correctness is mean@8 in this paper (called pass@1 in the hosted report). Raw distinct@8 counts verified keys across eight draws, including zero for unsolved prompts. For prompt x , let c_x be its number of correct draws and n_{xc} its count for correct key c . The within-cell correct-pair collision estimate is

$$\hat{C} = \frac{\sum_x \sum_c \binom{n_{xc}}{2}}{\sum_x \binom{c_x}{2}}.$$

It describes concentration conditional on successful draws, with prompts weighted by their number of correct pairs. Level summaries average the five domain estimates equally. When the certified correct-key count m_x is available, a uniform-correct-key reference replaces the numerator by $\sum_x \binom{c_x}{2} / m_x$. This differs from a uniform-action baseline and is a descriptive reference, not a normative sampling requirement. Countdown’s verifier accepts modes beyond its enumerated binary-expression library, including unary negation, so its total support is unknown and this reference is omitted. Intervals use 2,000 whole-prompt bootstrap replicates, seed 20260911, resampling within each domain and level and keeping all eight draws together. They are pointwise descriptive intervals.

Table 13: **GPT-5.6 Sol under strict executable grading.** Every cell contains 128 prompts and 1,024 responses. M and P are mean@8 and pass@8 (%); D is distinct@8. C is correct-pair collision (%), with 95% prompt-bootstrap interval); U is the uniform-correct-key collision reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	86.1	98.4	1.77	71.2 [66.1, 75.9]	–
1	Python	40.4	90.6	1.11	81.8 [76.9, 87.2]	0.8
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	65.4	86.7	1.57	64.7 [59.0, 70.6]	–
2	Python	8.5	11.7	0.12	100.0 [100.0, 100.0]	0.7
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	55.3	96.1	1.72	63.6 [56.8, 70.3]	6.9
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	71.2	89.8	1.38	76.4 [71.1, 81.8]	–
3	Python	9.4	14.1	0.14	100.0 [100.0, 100.0]	0.9
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	65.3	100.0	2.22	51.0 [45.3, 57.3]	6.8

Formatting sensitivity. Strict grades remain primary. A secondary typography normalizer was introduced after inspecting the first 15 responses. It fixes a frozen set of presentation conventions, including escaped Python/LaTeX syntax, and reuses the original executable validators without repairing calculations or program logic. It recovers 2,580 additional verified outputs; strict successes

and their keys are retained. This is a post hoc diagnostic, not a replacement benchmark. All 3,072 primary Python responses passed an independent serial recheck with no grade changes. One cold-start timeout in the secondary grading cache was reconciled with repeated serial verification; the original cache and correction audit remain preserved.

Table 14: **Post hoc formatting-normalized diagnostic.** Same samples and metric definitions as Table 13, with fixed presentation normalization before executable verification. The strict table remains primary; normalization was designed after the first 15 responses.

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	99.9	100.0	1.93	67.9 [63.3, 72.4]	–
1	Python	92.4	100.0	1.41	85.6 [82.3, 88.8]	1.4
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	98.5	100.0	1.99	63.9 [59.5, 68.4]	–
2	Python	60.7	99.2	1.12	94.8 [91.9, 97.2]	1.1
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	71.3	100.0	1.96	64.0 [58.1, 69.6]	6.8
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	99.0	100.0	1.70	75.7 [71.2, 80.1]	–
3	Python	56.5	98.4	1.09	94.3 [90.8, 97.2]	1.0
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	75.2	100.0	2.35	53.0 [47.3, 58.9]	6.7

Concentration and level. The strict five-domain collision average is 74.0%, 82.3% and 84.5% at Levels 1–3; the normalized diagnostic gives 74.1%, 81.2% and 83.6%. This aggregate pattern is not a monotonic law within domains. Graph collision is 89.7%, 83.3%, then 95.1%; normalized PantryPlan collision is 46.9%, 64.0%, then 53.0%. Normalized overall correctness also rebounds from 83.2% at Level 2 to 85.7% at Level 3. Thus a higher level label is not equivalent to a measured increase in difficulty for this model, and raw distinct-mode counts also change with correctness and support.

Level-3 Graph separates concentration from failed responses particularly clearly: all 1,024 outputs verify, eight draws produce 1.16 distinct keys on average, and 3,407 of 3,584 correct pairs share a key (95.1%). The uniform-correct-key reference is 19.0%; the excess is 76.1 percentage points (95% interval [73.5, 78.3]). Graph was selected after the GPT run for this illustration because its correctness is high, support is certified and formatting normalization changes no scores; both tables report all domains.

Interface and claim limits. Level-2/Level-3 MathIR instructions prescribe moving the right-side variable term left, removing the constant and then dividing. Python instructions suggest a nested small-divisor conditional chain, and PantryPlan instructions prefer particular ingredient families. These preferences can concentrate the very keys being measured. In addition, Level-1 PantryPlan requests a support mask whose quantities are supplied by a trusted projection; Levels 2–3 ask the model to provide quantities. MathIR verifies symbolic action compliance, which can reject a numerically valid shortcut that omits a required symbolic step. Neither these failures nor the cross-level differences isolate numerical reasoning ability. Canonical keys describe executed outputs, not hidden reasoning strategies. Eight draws cannot establish that an unseen mode has zero probability. The evidence supports concentration under the specified prompts and deployment configuration; it identifies neither a training cause nor a causal difficulty or model-size effect.

Reproduction record. The source record is `artifacts/frontier_modebench_gpt56sol_20260911/summary.json`. Its hash, complete cell statistics and run settings are copied into `results/frontier_hosted_20260911.json`; adjacent generated table bodies supply the displayed values. The evidence directory also retains `completion_audit.json`, `prompt_contract_`

578 audit.json, the original requests and responses, and interpretation_notes.md. Hosted-model
 579 results are separate from the training-checkpoint archive and its matched-seed comparisons.

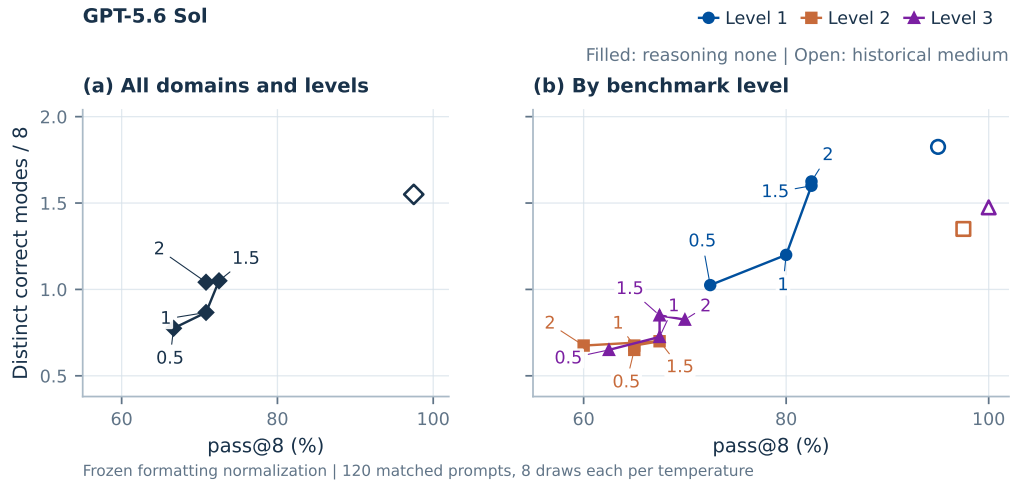


Figure 17: **Temperature traces pass@8 and verified breadth.** GPT-5.6 Sol, with 120 fixed prompts and eight draws each at four temperatures. Left: all domains and levels; right: separate levels, averaging five domains equally. Empirical pass@8 is the fraction with at least one correct draw; distinct@8 averages unique correct modes, including zeroes. Connected points use reasoning none and the frozen formatting normalizer. Open markers show the historical medium-reasoning reference on the same prompts (returned $T = 1$), unconnected because its configuration differs. This does not test temperature robustness with medium reasoning. Appendix H.3 gives pointwise uncertainty intervals for every level and all four measured temperatures.

H Comparison across Hosted Deployments

Figure 7 compares verified output breadth across hosted deployments. Its main display applies the frozen formatting normalizer to every deployment and uses the complete revised-wording Opus 5 Python cohort at all three levels. Every response enters its accuracy and mode-count metrics; outputs are not selected by correctness or provider outcome. The other cells use their original prompts. This is a descriptive comparison across the stated task formulations, rather than a common-prompt model ranking.

The tables and Graph figure below preserve every original-protocol result, including failed draws and native refusals. The separate Python comparison reports the original and revised conditions under both grading rules. These records expose the effect of the prompt change without replacing the original evidence. Level 3 remains an exploratory test extension; no hosted-model training or replay intervention is performed. Original deployments enter the comparison only with a passing integrity audit and all 15,360 responses.

Completed hosted comparison. The admitted deployments are GPT-5.6 Sol, Claude Opus 5, GPT-5.4, Grok 4.3, Kimi K3, Claude Opus 4.8, DeepSeek V4 Pro: 7 complete runs, 107,520 responses, and the same 128 held-out prompts per domain and level with eight stateless samples per prompt. Only runs with a passing completion audit and the identical frozen prompt digest enter these tables. No training, model-side tools or conversation history is added.

Provider configurations. GPT-5.6 Sol: hosted API, requested reasoning setting medium, native output limit 8192. Claude Opus 5: Anthropic Messages, requested reasoning setting medium (requested, provider-specific), native output limit 8192. GPT-5.4: hosted API, requested reasoning setting medium, native output limit 8192. Grok 4.3: hosted API, requested reasoning setting medium, native output limit 8192. Kimi K3: hosted API, requested reasoning setting medium, native output limit 8192. Claude Opus 4.8: Anthropic Messages, requested reasoning setting medium (requested, provider-specific), native output limit 8192. DeepSeek V4 Pro: hosted API, requested reasoning setting medium, native output limit 8192.

Provider effort labels do not establish equal compute, and native token accounting differs. The saved native request and returned usage determine each configuration; provider defaults remain unspecified whenever native responses omit them.

The appendix retains strict executable grades alongside formatting-normalized scores. The main display uses the same frozen formatting rule for every cell and the complete separately collected revised Opus 5 Python condition; all other displayed cells retain the original request wording. The shared typography diagnostic was developed after the first 15 GPT-5.6 Sol responses and frozen before the prospective comparison runs. It retains its post hoc status for GPT-5.6 Sol. Inherited route preferences and the Level-1 PantryPlan interface change limit cross-level interpretation. Countdown has no certified finite-total-support uniform reference. These data establish neither a training cause nor a causal model-size or difficulty effect. All pair-collision intervals are pointwise; overlapping or nonoverlapping cell intervals are not a paired model-difference test.

Collection interruption: DeepSeek V4 Pro. 13 HTTP-200 bodies had no terminal finish reason or visible answer and were preserved as provider protocol failures. A later storage interruption left 192 logged physical request starts without a saved outcome or usage; their outcome and possible billing remain unknown. The audit registers these attempts separately from the 15,360 terminal draws. Only missing terminal draws were recollected; saved terminal responses, including truncations, were retained. Reported token totals are subtotals of known usage, not zero-cost claims for failed or interrupted attempts. Source-bound recovery sidecars preserve the original receipts and the exact interruption accounting.

H.1 A separate Python prompt sensitivity test

After the original Opus 5 Python refusals, synthetic tasks were used to select a direct arithmetic-expression formulation with no system message. We then collected eight independent draws for every original Python test task: all 384 prompts across three levels, with unchanged adaptive thinking, medium effort, 8,192 output tokens, allowed operators, public inputs and frozen graders. The revised

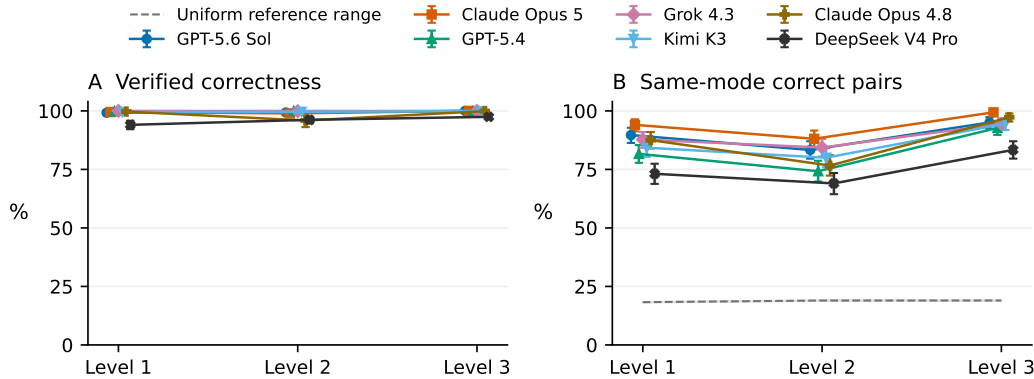


Figure 18: **Verified correctness and concentration in hosted Graph outputs.** Each deployment receives the same 128 prompts per level, with eight stateless samples per prompt. Intervals resample whole prompts. The gray band spans the models’ uniform-correct-key references, conditional on their observed correct draws. Graph was selected after the GPT-5.6 Sol run and before the prospective comparison runs. Level labels do not guarantee increasing difficulty for each model.

631 user instruction asks for a one-line boxed Python lambda returning any proper divisor for each listed
632 input. It does not prescribe a preferred divisor. Both task wording and system-message presence
633 change; collection time also differs. This exploratory follow-up therefore supports a practical prompt
634 repair without isolating its cause. Original refusals and all three residual refusals are retained. The
635 original-protocol appendix tables retain all original Python responses; the main display uses this
636 complete revised condition with an explicit caption note.

Table 15: **Opus 5 Python: original versus plain task wording.** Each row has 1,024 responses. R counts native refusals; A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). Strict and normalized results use the same frozen grader and typography rule. Original Level-2/3 collision is undefined because no prompt has two correct draws.

Condition	L	R	A	Strict		Normalized		
				D	C	A	D	C
Original	1	409	38.1	0.91	93.0	60.0	1.12	90.4
	2	1021	0.3	0.02	–	0.3	0.02	–
	3	1024	0.0	0.00	–	0.0	0.00	–
Plain user, no system	1	0	74.9	1.42	80.2	100.0	1.49	79.9
	2	0	71.8	1.31	84.7	100.0	1.38	85.7
	3	3	76.2	1.38	82.7	99.7	1.41	84.1

637 The new condition has no API failures or token-limit truncations. Strict grading accepts 2,282
638 responses; the frozen rule rescues 787 more by converting escaped percent signs (%) to Python
639 modulo (%). Thus every non-refused answer passes after this formatting-only correction. Normalized
640 collision is 79.9/85.7/84.1%, versus conditional uniform references of 1.43/1.15/1.15%. The residual
641 three refusals occur on two prompts whose other identical requests are accepted. No retry-until-
642 accepted selection is used.

643 The paired analysis resamples whole eight-draw prompt groups across conditions (20,000 replicates).
644 Normalized accuracy gains at Levels 1/2/3 are 40.0 [34.7, 45.5], 99.7 [99.3, 100.0], and 99.7 [99.2,
645 100.0] percentage points (pointwise 95% intervals). Output draws themselves are independent, and
646 the different correct-pair populations preclude interpreting a raw collision difference as a matched-
647 population prompt effect.

648 The saved sensitivity record binds the original and revised native receipts, request payloads, strict
649 regrade audit and frozen normalization outputs.

Separate first-valid retry diagnostic. After completing the fixed 3,072-response revised-wording cohort, a separate condition retries only its three remaining refused slots, retaining the first verified-valid response. All three succeed on their first additional request, and all reproduce a mode already seen for that prompt. This uses 3,075 requests to select 3,072 valid responses; its 100% accepted-set accuracy follows from selection and is not unconditional model accuracy. The selected set’s `distinct@8` is unchanged at 1.492, 1.375, 1.414 at Levels 1–3. Level-3 correct-pair collision changes from 84.0909% to 84.1518%; the other levels are unchanged. All initial and retry responses remain preserved. The main figure continues to use the complete fixed eight-draw condition, including its three refusals; these selected-validity results are only a separate diagnostic with variable request effort.

660 H.2 A matched requested-temperature sensitivity test

A separate fixed subset contains eight prompts from each of the five domains at each of three levels: 120 prompts, with eight independent draws per prompt and temperature. Grok 4.3 and Kimi K3 each receive requested temperatures $T = 1.0$ and $T = 1.5$, giving 960 draws per condition and 3,840 in total. The two conditions share task rows and all non-temperature request settings; both temperatures are collected concurrently within each deployment. Every response is retained. This subset is a separate experiment and does not replace any full-cohort result in the main hosted display.

Provider API contracts support the requested controls and the deployments accept the requests, but native responses do not report effective sampling temperature. DeepSeek is excluded from this contrast because its documented thinking-mode interface ignores the temperature setting. These contrasts therefore concern the tested requested settings, not arbitrary temperatures or all deployments.

Tables average all five domains equally within each level; All equally averages all 15 domain–level cells. Confidence intervals use 20,000 paired whole-prompt bootstrap resamples stratified by domain and level, retaining each eight-draw group. They are pointwise and have no multiplicity adjustment. Correct-pair collision remains conditional on correct draws and can change its eligible prompt population; an undefined cell makes the corresponding macro undefined.

Observed sensitivity. For Grok, the observed breadth change is +0.050 [-0.058, +0.158] modes, and the collision change is -0.78 [-3.86, +2.12] percentage points. These intervals do not show a clear broadening at the tested setting. The per-response accuracy change is -2.29 [-3.85, -0.94] percentage points. This does not establish invariance to temperature in general.

Kimi returns 696/960 token-limited outputs at $T = 1.5$, compared with 0 at $T = 1.0$. Strict accuracy falls from 96.56% to 21.25%, while raw breadth falls from 1.933 to 0.983 modes. This is a severe token-limit and accuracy failure, not a clean increase in correct-output concentration. The five-domain collision mean is undefined at $T = 1.5$ because some cells have no eligible correct pairs. Neither model has provider-declared refusals in this experiment.

† marks an omitted interval when fewer than all 20,000 bootstrap replicates define the contrast. For Kimi collision, only 11,815 and 11,616 replicates are defined at Levels 1 and 2; the raw conditional percentile summaries remain in the source record. We do not present them as ordinary 95% intervals. Undefined cells are never replaced by zero or silently omitted.

Full domain–level estimates, counts, native outcomes and collision eligibility remain in the source-bound temperature analysis. Countdown has no finite-total-support uniform reference; finite-support sensitivity summaries use the other four domains. The paper record authenticates every included condition summary, native completion audit, frozen normalizer, paired analysis source and sampling-control review, with the complete fixed response cohort retained throughout.

694 H.3 GPT-5.6 Sol: temperature, pass@8 and verified modes

Figure 17 plots empirical `pass@8` against `distinct@8` for $T \in \{0.5, 1.0, 1.5, 2.0\}$, with reasoning none. Each condition retains the same 120 held-out prompts (eight per domain–level cell), with eight stateless responses per prompt: 960 responses per temperature and 3,840 in total. A prompt contributes one to `pass@8` if at least one of its eight answers is verified correct, and contributes its number of distinct correct canonical keys to `distinct@8`. Zero-success groups and all unsuccessful

Table 16: **Requested-temperature sensitivity: Strict executable grades.** A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). D differences are mode counts; A/C differences are percentage points. Brackets give paired pointwise 95% intervals for the paired change in requested temperature. Every fixed draw remains in its original denominator.

Deployment	Level	Metric	$T = 1.0$	$T = 1.5$	Difference [95% interval]
Grok 4.3	1	A	98.12	95.62	-2.50 [-4.06, -0.94]
		D	1.900	2.000	+0.100 [-0.100, +0.300]
		C	72.89	73.16	+0.27 [-6.31, +6.37]
	2	A	99.06	99.38	+0.31 [-0.94, +1.56]
		D	1.425	1.425	+0.000 [-0.175, +0.200]
		C	86.61	85.20	-1.41 [-6.26, +3.30]
	3	A	98.75	94.06	-4.69 [-9.06, -0.94]
		D	1.500	1.550	+0.050 [-0.125, +0.225]
		C	84.04	82.85	-1.20 [-5.48, +2.79]
	All	A	98.65	96.35	-2.29 [-3.85, -0.94]
		D	1.608	1.658	+0.050 [-0.058, +0.158]
		C	81.18	80.40	-0.78 [-3.86, +2.12]
Kimi K3	1	A	92.50	22.50	-70.00 [-75.62, -63.75]
		D	2.200	1.075	-1.125 [-1.475, -0.775]
		C	66.23	53.12	-13.11 [-] [†]
	2	A	98.12	20.31	-77.81 [-82.50, -72.81]
		D	1.850	0.900	-0.950 [-1.300, -0.625]
		C	72.18	66.55	-5.63 [-] [†]
	3	A	99.06	20.94	-78.12 [-83.75, -72.81]
		D	1.750	0.975	-0.775 [-1.025, -0.525]
		C	78.33	—	—
	All	A	96.56	21.25	-75.31 [-78.44, -72.19]
		D	1.933	0.983	-0.950 [-1.133, -0.767]
		C	72.25	—	—

Table 17: **Requested-temperature sensitivity: Frozen formatting-normalized grades.** A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). D differences are mode counts; A/C differences are percentage points. Brackets give paired pointwise 95% intervals for the paired change in requested temperature. Every fixed draw remains in its original denominator.

Deployment	Level	Metric	$T = 1.0$	$T = 1.5$	Difference [95% interval]
Grok 4.3	1	A	98.12	95.62	-2.50 [-4.06, -0.94]
		D	1.900	2.000	+0.100 [-0.100, +0.300]
		C	72.89	73.16	+0.27 [-6.31, +6.37]
	2	A	99.06	99.38	+0.31 [-0.94, +1.56]
		D	1.425	1.425	+0.000 [-0.175, +0.200]
		C	86.61	85.20	-1.41 [-6.26, +3.30]
	3	A	98.75	94.06	-4.69 [-9.06, -0.94]
		D	1.500	1.550	+0.050 [-0.125, +0.225]
		C	84.04	82.85	-1.20 [-5.48, +2.79]
	All	A	98.65	96.35	-2.29 [-3.85, -0.94]
		D	1.608	1.658	+0.050 [-0.058, +0.158]
		C	81.18	80.40	-0.78 [-3.86, +2.12]
Kimi K3	1	A	92.50	22.81	-69.69 [-75.31, -63.44]
		D	2.200	1.075	-1.125 [-1.475, -0.775]
		C	66.23	53.61	-12.62 [-] [†]
	2	A	98.12	20.31	-77.81 [-82.50, -72.81]
		D	1.850	0.900	-0.950 [-1.300, -0.625]
		C	72.18	66.55	-5.63 [-] [†]
	3	A	99.38	20.94	-78.44 [-84.06, -73.12]
		D	1.750	0.975	-0.775 [-1.025, -0.525]
		C	78.18	—	—
	All	A	96.67	21.35	-75.31 [-78.44, -72.19]
		D	1.933	0.983	-0.950 [-1.133, -0.767]
		C	72.20	—	—

draws remain in the denominator. We compute success directly from the saved groups, not as $1 - (1 - p)^8$ from pooled per-response accuracy; task difficulty varies across prompts.

The deployment rejected non-default temperatures with medium reasoning. The connected curves therefore use the supported reasoning none condition. The separate historical medium reference uses the same 120 prompts and eight draws; its requests omitted temperature and receipts returned $T = 1$. It remains unconnected because both reasoning setting and collection time differ. This does not test temperature robustness of the medium condition.

Each level averages five domains equally; All averages all 15 cells. Pointwise 95 percent intervals use 20,000 paired whole-prompt bootstrap resamples stratified by domain and level, retaining all eight draws. There is no multiplicity adjustment. A zero-width empirical interval reflects constant observed groups, not certain success on unseen prompts.

Table 18: **GPT temperature frontier: Strict executable grades.** P is solved prompts (% , pass@8); D is distinct@8; A is per-response accuracy (%), retained as a separate diagnostic. Brackets give pointwise 95% intervals. M is the historical medium reference; numeric temperatures all use reasoning none and the same fixed eight-draw sampling budget.

T	Level	P [95% interval]	D [95% interval]	A [95% interval]
0.5	1	62.5 [50.0, 75.0]	0.925 [0.675, 1.200]	52.5 [41.2, 63.4]
	2	45.0 [37.5, 55.0]	0.450 [0.375, 0.550]	29.7 [22.5, 36.2]
	3	47.5 [35.0, 60.0]	0.475 [0.350, 0.600]	40.0 [27.8, 52.2]
	All	51.7 [45.0, 58.3]	0.617 [0.517, 0.717]	40.7 [34.8, 46.7]
1.0	1	77.5 [65.0, 90.0]	1.175 [0.950, 1.400]	53.1 [42.2, 63.7]
	2	45.0 [35.0, 55.0]	0.475 [0.350, 0.600]	29.4 [22.2, 36.6]
	3	52.5 [40.0, 65.0]	0.550 [0.400, 0.700]	38.8 [27.8, 50.0]
	All	58.3 [51.7, 65.0]	0.733 [0.642, 0.833]	40.4 [34.7, 46.0]
1.5	1	82.5 [70.0, 92.5]	1.525 [1.250, 1.825]	54.4 [44.1, 64.7]
	2	47.5 [37.5, 57.5]	0.475 [0.375, 0.575]	27.8 [22.2, 33.1]
	3	55.0 [42.5, 67.5]	0.675 [0.475, 0.875]	42.2 [31.2, 53.4]
	All	61.7 [55.0, 68.3]	0.892 [0.775, 1.008]	41.5 [36.0, 46.8]
2.0	1	82.5 [70.0, 92.5]	1.600 [1.325, 1.900]	58.4 [48.1, 68.4]
	2	40.0 [32.5, 47.5]	0.425 [0.350, 0.500]	26.6 [21.2, 31.9]
	3	55.0 [42.5, 67.5]	0.625 [0.475, 0.800]	34.7 [25.6, 44.4]
	All	59.2 [53.3, 65.0]	0.883 [0.767, 1.000]	39.9 [34.9, 44.8]
M	1	95.0 [87.5, 100.0]	1.675 [1.425, 1.925]	79.4 [71.6, 86.6]
	2	75.0 [67.5, 80.0]	1.025 [0.825, 1.225]	59.4 [50.3, 67.2]
	3	85.0 [80.0, 92.5]	1.225 [1.000, 1.475]	71.2 [63.1, 79.7]
	All	85.0 [81.7, 88.3]	1.308 [1.183, 1.442]	70.0 [65.2, 74.6]

Observed temperature frontier. From $T = 0.5$ to $T = 1.5$, normalized pass@8 rises from 66.67% to 72.50%, while distinct@8 rises from 0.775 to 1.050. At $T = 2.0$, the corresponding values are 70.83% and 1.042. Thus $T = 1.5$ is the best observed aggregate point on both axes among these four temperatures. This finite exploratory sweep does not establish a population optimum or a continuous Pareto boundary, and level-specific curves differ. In particular, Level 1 continues gaining modes at the highest temperature even though the overall point declines slightly.

Per-response accuracy nevertheless decreases from 55.52% to 50.21% between the endpoints. That diagnostic is distinct from pass@8: success within eight draws can improve while an individual draw becomes less reliable on these same prompts.

The paired $T = 2.0$ minus $T = 0.5$ contrast is 4.2 [-0.8, 9.2] pass@8 percentage points and 0.267 [0.183, 0.350] distinct correct modes, with pointwise 95% intervals. All prompt-level records, domain estimates, original per-response and collision analyses, and paired comparisons remain in the retained reports for both strict and normalized grading.

Retained outcomes. At $T = 0.5$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 1.0$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 1.5$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 2.0$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. These outcome counts retain every

Table 19: **GPT temperature frontier: Frozen formatting-normalized grades.** P is solved prompts (% pass@8); D is distinct@8; A is per-response accuracy (%), retained as a separate diagnostic. Brackets give pointwise 95% intervals. M is the historical medium reference; numeric temperatures all use reasoning none and the same fixed eight-draw sampling budget.

<i>T</i>	Level	P [95% interval]	D [95% interval]	A [95% interval]
0.5	1	72.5 [60.0, 85.0]	1.025 [0.775, 1.300]	67.8 [55.9, 79.1]
	2	65.0 [57.5, 75.0]	0.650 [0.575, 0.750]	46.2 [38.1, 53.8]
	3	62.5 [50.0, 75.0]	0.650 [0.525, 0.775]	52.5 [40.3, 64.4]
	All	66.7 [60.0, 73.3]	0.775 [0.683, 0.875]	55.5 [49.3, 61.7]
1.0	1	80.0 [67.5, 90.0]	1.200 [0.975, 1.425]	67.2 [55.3, 78.4]
	2	65.0 [55.0, 75.0]	0.675 [0.550, 0.800]	44.4 [36.6, 52.2]
	3	67.5 [55.0, 80.0]	0.725 [0.600, 0.875]	50.0 [39.4, 60.9]
	All	70.8 [64.2, 77.5]	0.867 [0.775, 0.967]	53.9 [47.9, 59.6]
1.5	1	82.5 [70.0, 92.5]	1.600 [1.325, 1.900]	67.2 [55.9, 78.1]
	2	67.5 [57.5, 77.5]	0.700 [0.600, 0.800]	39.1 [32.5, 45.6]
	3	67.5 [55.0, 80.0]	0.850 [0.650, 1.050]	50.9 [40.0, 61.9]
	All	72.5 [65.8, 79.2]	1.050 [0.925, 1.175]	52.4 [46.8, 57.9]
2.0	1	82.5 [70.0, 92.5]	1.625 [1.325, 1.925]	68.4 [57.5, 78.8]
	2	60.0 [52.5, 67.5]	0.675 [0.575, 0.775]	37.5 [31.6, 43.1]
	3	70.0 [57.5, 82.5]	0.825 [0.650, 1.000]	44.7 [34.7, 54.7]
	All	70.8 [65.0, 76.7]	1.042 [0.925, 1.158]	50.2 [44.9, 55.3]
M	1	95.0 [87.5, 100.0]	1.825 [1.575, 2.075]	93.4 [86.6, 98.8]
	2	97.5 [92.5, 100.0]	1.350 [1.175, 1.550]	80.9 [73.8, 87.5]
	3	100.0 [100.0, 100.0]	1.475 [1.250, 1.700]	85.0 [80.0, 90.0]
	All	97.5 [95.0, 100.0]	1.550 [1.425, 1.683]	86.5 [82.8, 89.9]

registered draw and are separate from the verifier-based correctness and canonical-key measurements above.

H.4 Native provider outcomes and undefined concentration

Provider-declared refusals and filtering are deployment outcomes, not evidence that the underlying model cannot solve the mathematical task. Empty refusals remain failed draws under the frozen verifier; no requests or grades are overwritten. Classification uses native metadata, not a semantic detector of refusals written as ordinary answer text.

Correct-pair collision is undefined when no prompt supplies two correct draws. The five-domain mean is therefore undefined if any domain has no eligible pairs; such entries remain – rather than zero or a mean over fewer domains.

GPT-5.6 Sol native completion states: completed: 15,360. Empty visible answers are reported separately from provider refusals.

Claude Opus 5 native completion states: end_turn: 12,838, refusal: 2,522. Empty visible answers are reported separately from provider refusals.

Claude Opus 5 returns provider-declared refusals for 409, 1021, 1024 of the 1,024 Python requests at Levels 1, 2 and 3, respectively. The native refusal categories on these Python draws are cyber: 2454. These are benign factor-selection tasks; the reported deployment policy must be distinguished from concentration among correct Python outputs.

GPT-5.4 native completion states: completed: 15,355, incomplete: 5. Empty visible answers are reported separately from provider refusals.

Grok 4.3 native completion states: stop: 15,360. Empty visible answers are reported separately from provider refusals.

Kimi K3 native completion states: length: 15, stop: 15,345. Empty visible answers are reported separately from provider refusals.

Claude Opus 4.8 native completion states: end_turn: 15,355, max_tokens: 5. Empty visible answers are reported separately from provider refusals.

Table 20: **Native service outcomes: GPT-5.6 Sol.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 21: **Native service outcomes: Claude Opus 5.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	409	0	409
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	1021	0	1021
2	MathIR	35	0	35
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	1024	0	1024
3	MathIR	33	0	32
3	PantryPlan	0	0	0

754 Claude Opus 4.8 returns provider-declared refusals for 0, 0, 0 of the 1,024 Python requests at Levels
755 1, 2 and 3, respectively. The native refusal categories on these Python draws are not supplied.
756 These are benign factor-selection tasks; the reported deployment policy must be distinguished from
757 concentration among correct Python outputs.

758 DeepSeek V4 Pro native completion states: length: 632, stop: 14,728. Empty visible answers are
759 reported separately from provider refusals.

760 The result record `results/frontier_comparison_20260911.json` binds every source summary,
761 completion audit, frozen prompt digest, raw-sample digest and primary graded sample digest. Pending
762 deployments are omitted from both the figure and numerical comparisons. The figure was selected
763 after the reference GPT run for its high strict correctness, certified support and lack of formatting
764 corrections, and before collecting the prospective comparison runs. All five domains are retained in
765 the tables, including disagreement across models.

Table 22: **Native service outcomes: GPT-5.4.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	1
2	Graph	0	0	0
2	Countdown	0	0	1
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	1
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	2

Table 23: **Native service outcomes: Grok 4.3.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 24: **Native service outcomes: Kimi K3.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 25: **Native service outcomes: Claude Opus 4.8.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	3
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	1

Table 26: **Native service outcomes: DeepSeek V4 Pro.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	59
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	4
1	PantryPlan	0	0	13
2	Graph	0	0	38
2	Countdown	0	0	1
2	Python	0	0	0
2	MathIR	0	0	59
2	PantryPlan	0	0	80
3	Graph	0	0	24
3	Countdown	0	0	7
3	Python	0	0	0
3	MathIR	0	0	100
3	PantryPlan	0	0	246

Table 27: **Strict level summaries for completed hosted deployments.** Each level weights the five domains equally. M and P are mean@8 and pass@8 (%); D is raw distinct@8. C is correct-pair collision (%; with pointwise 95% whole-prompt intervals), pair weighted within each domain. Provider settings differ; these are descriptive outputs.

Deployment	Level	M	P	D	C (95% interval)
GPT-5.6 Sol	1	83.0	95.8	1.71	74.0 [71.9, 76.1]
GPT-5.6 Sol	2	62.9	77.5	1.16	82.3 [80.3, 84.4]
GPT-5.6 Sol	3	68.7	80.3	1.18	84.5 [82.8, 86.2]
Claude Opus 5	1	85.3	94.5	1.32	86.3 [84.5, 88.0]
Claude Opus 5	2	76.6	80.2	1.16	—
Claude Opus 5	3	77.8	79.8	1.07	—
GPT-5.4	1	93.7	98.1	2.09	67.2 [65.1, 69.3]
GPT-5.4	2	80.6	97.7	1.61	74.7 [72.4, 76.8]
GPT-5.4	3	83.3	97.8	1.57	78.8 [76.9, 80.7]
Grok 4.3	1	98.3	99.5	1.97	72.4 [70.7, 74.0]
Grok 4.3	2	99.3	100.0	1.57	82.1 [80.6, 83.7]
Grok 4.3	3	98.8	100.0	1.55	83.0 [81.5, 84.6]
Kimi K3	1	96.9	98.4	2.31	62.5 [60.6, 64.4]
Kimi K3	2	96.8	100.0	2.05	69.4 [67.6, 71.2]
Kimi K3	3	97.8	100.0	1.90	74.6 [73.1, 76.2]
Claude Opus 4.8	1	96.4	98.8	1.64	78.4 [76.6, 80.3]
Claude Opus 4.8	2	98.5	100.0	1.67	77.2 [75.4, 79.1]
Claude Opus 4.8	3	98.9	99.8	1.51	83.4 [81.8, 84.9]
DeepSeek V4 Pro	1	96.4	98.4	2.01	71.0 [69.4, 72.7]
DeepSeek V4 Pro	2	93.7	99.1	1.84	68.7 [67.0, 70.3]
DeepSeek V4 Pro	3	91.0	99.7	1.62	78.8 [76.9, 80.9]

Table 28: **GPT-5.6 Sol: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	86.1	98.4	1.77	71.2 [66.1, 75.9]	–
1	Python	40.4	90.6	1.11	81.8 [76.9, 87.2]	0.8
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	65.4	86.7	1.57	64.7 [59.0, 70.6]	–
2	Python	8.5	11.7	0.12	100.0 [100.0, 100.0]	0.7
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	55.3	96.1	1.72	63.6 [56.8, 70.3]	6.9
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	71.2	89.8	1.38	76.4 [71.1, 81.8]	–
3	Python	9.4	14.1	0.14	100.0 [100.0, 100.0]	0.9
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	65.3	100.0	2.22	51.0 [45.3, 57.3]	6.8

Table 29: **Claude Opus 5: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.15	94.0 [91.3, 96.4]	18.3
1	Countdown	100.0	100.0	1.59	78.8 [74.6, 83.0]	–
1	Python	38.1	81.2	0.91	93.0 [88.7, 96.6]	0.7
1	MathIR	90.5	91.4	1.09	92.2 [88.8, 95.4]	20.0
1	PantryPlan	98.4	100.0	1.86	73.5 [68.5, 78.2]	6.5
2	Graph	98.9	100.0	1.32	88.1 [84.5, 91.6]	18.9
2	Countdown	100.0	100.0	1.81	69.3 [65.2, 73.6]	–
2	Python	0.3	2.3	0.02	–	–
2	MathIR	89.1	98.4	1.06	96.5 [93.9, 98.6]	20.0
2	PantryPlan	94.5	100.0	1.56	80.7 [76.0, 85.3]	6.4
3	Graph	100.0	100.0	1.02	99.4 [98.3, 100.0]	19.0
3	Countdown	99.9	100.0	1.66	76.2 [71.7, 80.7]	–
3	Python	0.0	0.0	0.00	–	–
3	MathIR	95.3	99.2	1.02	98.8 [97.5, 99.8]	20.0
3	PantryPlan	93.6	100.0	1.63	78.8 [74.2, 83.4]	6.5

Table 30: **GPT-5.4: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.8	100.0	1.57	81.6 [77.8, 85.4]	18.3
1	Countdown	91.8	100.0	1.73	76.6 [71.2, 81.3]	–
1	Python	87.6	100.0	2.52	56.4 [51.3, 61.5]	1.4
1	MathIR	89.3	90.6	1.45	78.6 [74.0, 83.4]	20.0
1	PantryPlan	99.9	100.0	3.16	42.8 [38.4, 47.4]	6.5
2	Graph	99.9	100.0	1.74	74.2 [69.8, 78.7]	19.0
2	Countdown	91.1	100.0	1.98	63.0 [58.6, 67.5]	–
2	Python	47.9	96.1	1.47	72.9 [64.6, 80.2]	1.0
2	MathIR	87.0	92.2	0.92	100.0 [100.0, 100.0]	20.0
2	PantryPlan	77.0	100.0	1.93	63.2 [58.4, 68.6]	6.8
3	Graph	99.8	100.0	1.23	92.7 [89.7, 95.4]	19.0
3	Countdown	82.2	93.8	1.72	69.1 [64.4, 73.5]	–
3	Python	50.1	97.7	1.58	74.4 [68.6, 79.8]	1.1
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	86.5	100.0	2.34	57.8 [52.8, 63.4]	6.6

Table 31: **Grok 4.3: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	100.0	100.0	1.40	87.8 [84.2, 91.0]	18.3
1	Countdown	99.9	100.0	2.05	64.8 [60.1, 69.5]	–
1	Python	99.8	100.0	1.41	88.0 [84.8, 91.0]	1.4
1	MathIR	92.4	97.7	1.24	90.1 [86.9, 93.1]	20.0
1	PantryPlan	99.4	100.0	3.77	31.4 [27.8, 35.1]	6.5
2	Graph	99.9	100.0	1.52	84.3 [80.5, 88.0]	19.0
2	Countdown	99.8	100.0	1.99	65.8 [61.8, 70.2]	–
2	Python	99.8	100.0	1.05	98.8 [97.8, 99.6]	1.1
2	MathIR	97.5	100.0	1.12	94.9 [92.4, 97.2]	20.0
2	PantryPlan	99.5	100.0	2.16	66.6 [61.2, 71.4]	6.5
3	Graph	100.0	100.0	1.17	94.5 [92.0, 96.8]	19.0
3	Countdown	99.9	100.0	1.98	66.3 [61.7, 71.0]	–
3	Python	99.9	100.0	1.04	98.6 [97.2, 99.6]	1.2
3	MathIR	95.0	100.0	1.05	97.8 [95.9, 99.3]	20.0
3	PantryPlan	99.2	100.0	2.52	58.0 [53.1, 63.5]	6.5

Table 32: **Kimi K3: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.48	84.3 [80.4, 87.8]	18.3
1	Countdown	99.1	100.0	1.91	68.4 [63.8, 73.3]	–
1	Python	97.9	100.0	2.52	57.7 [53.5, 61.9]	1.4
1	MathIR	90.2	92.2	1.47	75.8 [70.7, 80.7]	20.0
1	PantryPlan	97.5	100.0	4.18	26.2 [23.1, 29.2]	6.5
2	Graph	99.6	100.0	1.62	80.0 [76.0, 84.1]	19.0
2	Countdown	98.4	100.0	1.98	64.4 [59.9, 68.7]	–
2	Python	98.2	100.0	2.44	61.9 [57.6, 66.2]	1.1
2	MathIR	94.3	100.0	1.02	99.8 [99.3, 100.0]	20.0
2	PantryPlan	93.3	100.0	3.20	41.3 [36.6, 45.9]	6.6
3	Graph	99.9	100.0	1.19	94.6 [91.8, 96.9]	19.0
3	Countdown	97.5	100.0	1.84	70.6 [66.1, 74.8]	–
3	Python	98.0	100.0	2.34	64.7 [60.1, 69.0]	1.2
3	MathIR	98.1	100.0	1.00	100.0 [100.0, 100.0]	20.0
3	PantryPlan	95.6	100.0	3.13	43.4 [38.6, 48.3]	6.5

Table 33: **Claude Opus 4.8: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.7	100.0	1.34	87.5 [83.9, 91.0]	18.3
1	Countdown	95.6	99.2	1.78	69.8 [64.9, 74.5]	–
1	Python	98.0	100.0	1.40	85.7 [81.7, 89.4]	1.4
1	MathIR	92.3	94.5	1.10	93.7 [91.1, 96.4]	20.0
1	PantryPlan	96.6	100.0	2.59	55.3 [50.3, 60.1]	6.5
2	Graph	95.9	100.0	1.66	76.6 [72.3, 81.1]	18.8
2	Countdown	99.9	100.0	1.99	65.3 [60.9, 69.9]	–
2	Python	100.0	100.0	1.34	86.0 [82.3, 89.6]	1.2
2	MathIR	97.0	100.0	1.24	90.1 [86.7, 93.4]	20.0
2	PantryPlan	99.5	100.0	2.11	67.9 [62.8, 73.1]	6.5
3	Graph	99.9	100.0	1.08	97.3 [95.5, 98.9]	19.0
3	Countdown	97.9	99.2	1.80	72.5 [68.2, 76.6]	–
3	Python	98.9	100.0	1.32	86.4 [82.5, 90.1]	1.2
3	MathIR	99.1	100.0	1.03	99.1 [98.1, 99.8]	20.0
3	PantryPlan	98.5	100.0	2.33	61.6 [56.7, 66.6]	6.5

Table 34: **DeepSeek V4 Pro: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	94.0	100.0	1.80	73.1 [68.8, 77.4]	18.1
1	Countdown	100.0	100.0	2.02	65.8 [61.4, 70.3]	–
1	Python	99.3	100.0	1.41	88.8 [86.1, 91.4]	1.4
1	MathIR	90.1	92.2	1.11	93.9 [91.2, 96.2]	20.0
1	PantryPlan	98.5	100.0	3.73	33.5 [29.8, 37.2]	6.5
2	Graph	96.2	100.0	1.97	69.0 [64.4, 73.5]	19.1
2	Countdown	99.9	100.0	2.04	63.4 [59.2, 67.7]	–
2	Python	99.6	100.0	1.05	98.4 [97.1, 99.4]	1.2
2	MathIR	85.4	95.3	1.82	51.7 [49.8, 53.9]	20.0
2	PantryPlan	87.1	100.0	2.34	61.1 [56.0, 66.2]	6.5
3	Graph	97.5	100.0	1.52	83.4 [79.6, 87.0]	18.9
3	Countdown	99.2	100.0	1.98	67.5 [63.1, 72.2]	–
3	Python	99.6	100.0	1.08	97.8 [96.3, 99.0]	1.2
3	MathIR	88.5	100.0	1.44	83.6 [79.2, 87.6]	20.0
3	PantryPlan	70.3	98.4	2.08	61.9 [55.9, 68.2]	6.5

Table 35: **GPT-5.6 Sol: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	99.9	100.0	1.93	67.9 [63.3, 72.4]	–
1	Python	92.4	100.0	1.41	85.6 [82.3, 88.8]	1.4
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	98.5	100.0	1.99	63.9 [59.5, 68.4]	–
2	Python	60.7	99.2	1.12	94.8 [91.9, 97.2]	1.1
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	71.3	100.0	1.96	64.0 [58.1, 69.6]	6.8
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	99.0	100.0	1.70	75.7 [71.2, 80.1]	–
3	Python	56.5	98.4	1.09	94.3 [90.8, 97.2]	1.0
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	75.2	100.0	2.35	53.0 [47.3, 58.9]	6.7

Table 36: **Claude Opus 5: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.15	94.0 [91.3, 96.4]	18.3
1	Countdown	100.0	100.0	1.59	78.8 [74.6, 83.0]	–
1	Python	60.0	94.5	1.12	90.4 [86.2, 94.3]	0.9
1	MathIR	90.5	91.4	1.09	92.2 [88.8, 95.4]	20.0
1	PantryPlan	98.4	100.0	1.86	73.5 [68.5, 78.2]	6.5
2	Graph	98.9	100.0	1.32	88.1 [84.5, 91.6]	18.9
2	Countdown	100.0	100.0	1.81	69.3 [65.2, 73.6]	–
2	Python	0.3	2.3	0.02	–	–
2	MathIR	89.1	98.4	1.06	96.5 [93.9, 98.6]	20.0
2	PantryPlan	99.9	100.0	1.61	79.7 [75.0, 84.3]	6.5
3	Graph	100.0	100.0	1.02	99.4 [98.3, 100.0]	19.0
3	Countdown	100.0	100.0	1.66	76.1 [71.7, 80.7]	–
3	Python	0.0	0.0	0.00	–	–
3	MathIR	95.3	99.2	1.02	98.8 [97.5, 99.8]	20.0
3	PantryPlan	99.4	100.0	1.70	77.3 [72.6, 82.0]	6.5

Table 37: **GPT-5.4: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.8	100.0	1.57	81.6 [77.8, 85.4]	18.3
1	Countdown	99.7	100.0	1.80	73.6 [68.5, 78.3]	–
1	Python	98.7	100.0	2.57	56.8 [51.9, 61.6]	1.4
1	MathIR	89.3	90.6	1.45	78.6 [74.0, 83.4]	20.0
1	PantryPlan	99.9	100.0	3.16	42.8 [38.4, 47.4]	6.5
2	Graph	99.9	100.0	1.74	74.2 [69.8, 78.7]	19.0
2	Countdown	98.4	100.0	2.12	61.1 [57.0, 65.1]	–
2	Python	91.3	100.0	1.87	74.0 [69.4, 78.1]	1.1
2	MathIR	87.0	92.2	0.92	100.0 [100.0, 100.0]	20.0
2	PantryPlan	77.0	100.0	1.93	63.2 [58.4, 68.6]	6.8
3	Graph	99.8	100.0	1.23	92.7 [89.7, 95.4]	19.0
3	Countdown	99.0	100.0	1.96	68.4 [64.2, 72.7]	–
3	Python	91.6	100.0	1.91	72.6 [68.3, 77.1]	1.2
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	86.5	100.0	2.34	57.8 [52.8, 63.4]	6.6

Table 38: **Grok 4.3: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	100.0	100.0	1.40	87.8 [84.2, 91.0]	18.3
1	Countdown	99.9	100.0	2.05	64.8 [60.1, 69.5]	–
1	Python	99.8	100.0	1.41	88.0 [84.8, 91.0]	1.4
1	MathIR	92.4	97.7	1.24	90.1 [86.9, 93.1]	20.0
1	PantryPlan	99.4	100.0	3.77	31.4 [27.8, 35.1]	6.5
2	Graph	99.9	100.0	1.52	84.3 [80.5, 88.0]	19.0
2	Countdown	99.8	100.0	1.99	65.8 [61.8, 70.2]	–
2	Python	99.8	100.0	1.05	98.8 [97.8, 99.6]	1.1
2	MathIR	97.5	100.0	1.12	94.9 [92.4, 97.2]	20.0
2	PantryPlan	99.5	100.0	2.16	66.6 [61.2, 71.4]	6.5
3	Graph	100.0	100.0	1.17	94.5 [92.0, 96.8]	19.0
3	Countdown	99.9	100.0	1.98	66.3 [61.7, 71.0]	–
3	Python	99.9	100.0	1.04	98.6 [97.2, 99.6]	1.2
3	MathIR	95.0	100.0	1.05	97.8 [95.9, 99.3]	20.0
3	PantryPlan	99.2	100.0	2.52	58.0 [53.1, 63.5]	6.5

Table 39: **Kimi K3: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.48	84.3 [80.4, 87.8]	18.3
1	Countdown	99.4	100.0	1.91	68.4 [63.9, 73.3]	–
1	Python	98.2	100.0	2.53	57.7 [53.5, 61.9]	1.4
1	MathIR	90.2	92.2	1.47	75.8 [70.7, 80.7]	20.0
1	PantryPlan	97.5	100.0	4.18	26.2 [23.1, 29.2]	6.5
2	Graph	99.6	100.0	1.62	80.0 [76.0, 84.1]	19.0
2	Countdown	98.7	100.0	1.98	64.5 [60.1, 68.9]	–
2	Python	98.3	100.0	2.44	61.9 [57.6, 66.2]	1.1
2	MathIR	94.3	100.0	1.02	99.8 [99.3, 100.0]	20.0
2	PantryPlan	93.4	100.0	3.20	41.4 [36.7, 46.0]	6.6
3	Graph	99.9	100.0	1.19	94.6 [91.8, 96.9]	19.0
3	Countdown	98.1	100.0	1.84	70.3 [65.8, 74.5]	–
3	Python	98.5	100.0	2.34	64.9 [60.3, 69.2]	1.2
3	MathIR	98.1	100.0	1.00	100.0 [100.0, 100.0]	20.0
3	PantryPlan	95.6	100.0	3.13	43.4 [38.6, 48.3]	6.5

Table 40: **Claude Opus 4.8: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.7	100.0	1.34	87.5 [83.9, 91.0]	18.3
1	Countdown	95.6	99.2	1.78	69.8 [64.9, 74.5]	–
1	Python	98.6	100.0	1.40	85.8 [81.8, 89.5]	1.5
1	MathIR	92.3	94.5	1.10	93.7 [91.1, 96.4]	20.0
1	PantryPlan	96.6	100.0	2.59	55.3 [50.3, 60.1]	6.5
2	Graph	95.9	100.0	1.66	76.6 [72.3, 81.1]	18.8
2	Countdown	99.9	100.0	1.99	65.3 [60.9, 69.9]	–
2	Python	100.0	100.0	1.34	86.0 [82.3, 89.6]	1.2
2	MathIR	97.0	100.0	1.24	90.1 [86.7, 93.4]	20.0
2	PantryPlan	99.5	100.0	2.11	67.9 [62.8, 73.1]	6.5
3	Graph	99.9	100.0	1.08	97.3 [95.5, 98.9]	19.0
3	Countdown	97.9	99.2	1.80	72.5 [68.2, 76.6]	–
3	Python	98.9	100.0	1.32	86.4 [82.5, 90.1]	1.2
3	MathIR	99.1	100.0	1.03	99.1 [98.1, 99.8]	20.0
3	PantryPlan	98.7	100.0	2.33	61.7 [56.7, 66.6]	6.5

Table 41: **DeepSeek V4 Pro: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	94.0	100.0	1.80	73.1 [68.8, 77.4]	18.1
1	Countdown	100.0	100.0	2.02	65.8 [61.4, 70.3]	–
1	Python	99.4	100.0	1.41	88.8 [86.1, 91.4]	1.4
1	MathIR	90.1	92.2	1.11	93.9 [91.2, 96.2]	20.0
1	PantryPlan	98.5	100.0	3.73	33.5 [29.8, 37.2]	6.5
2	Graph	96.2	100.0	1.97	69.0 [64.4, 73.5]	19.1
2	Countdown	99.9	100.0	2.04	63.4 [59.2, 67.7]	–
2	Python	99.7	100.0	1.05	98.4 [97.1, 99.4]	1.2
2	MathIR	85.4	95.3	1.82	51.7 [49.8, 53.9]	20.0
2	PantryPlan	87.1	100.0	2.34	61.1 [56.0, 66.2]	6.5
3	Graph	97.5	100.0	1.52	83.4 [79.6, 87.0]	18.9
3	Countdown	99.2	100.0	1.98	67.5 [63.1, 72.2]	–
3	Python	99.7	100.0	1.08	97.8 [96.3, 99.0]	1.2
3	MathIR	88.5	100.0	1.44	83.6 [79.2, 87.6]	20.0
3	PantryPlan	70.3	98.4	2.08	61.9 [55.9, 68.2]	6.5

766 I Related Work

767 Repeated sampling and verifier-guided selection depend on the proposal distribution [Brown et al.,
768 2024, Snell et al., 2024]. Prior work studies diversity loss in alignment and reasoning [Kirk et al.,
769 2024, Dang et al., 2025], entropy dynamics [Cui et al., 2025], and high-budget coverage [Yue et al.,
770 2025]. Semantic clustering [Farquhar et al., 2024] addresses open-ended outputs; MODEBENCH
771 instead uses executable identity keys. Sampling-aware objectives promote useful proposal sets [Song
772 et al., 2025, Chen et al., 2025, Li et al., 2025, 2026a,b, Lochab et al., 2026]. MaxRL optimizes
773 best-of- k success [Tajwar et al., 2026], which equals $\text{pass}@k$ for binary rewards. Replay has a
774 long history in reinforcement and continual learning [Lin, 1992, Rolnick et al., 2019]; RLEP reuses
775 successful reasoning traces [Zhang et al., 2025]. Our bank retains one policy exemplar per verified key
776 and weights keys uniformly, connecting persistent success memory to novelty and quality-diversity
777 search [Lehman and Stanley, 2011, Mouret and Clune, 2015].

778 J Claims and Their Evidential Basis

779 The training and hosted studies share a measurement question: how is correct probability allocated
780 among verified alternatives? They supply different kinds of evidence. The table states the basis of
781 each central claim and the inference it supports; motivating downstream benefits remain hypotheses.

Claim	Identifiable basis	Scope of the inference
Success and breadth differ	Identity and concavity: Lemma A.1	At fixed correctness, concentration in majorization order lowers expected breadth. More correct samples can also raise the raw mode count.
Uniform categorical replay balances mass	Identity: Equation (4)	Likelihood replay already contains a balancing term. This neither requires a separate balance loss nor equates exemplar scores with key probabilities.
Binary reward can induce collapse	Conditional theorem: Theorem K.3	Independent canonical-outcome logits, common length normalization, Euclidean mean flow, and a unique initial winner; no neural-optimizer guarantee.
Replay remains available when fresh gradients vanish	Conditional score calculation: Appendix K.3	Equal-reward groups have zero centered task update; bank scoring need not. An available gradient does not ensure improvement after a shared-parameter update.
Persistent replay protects banked modes	Conditional theorem and corollary: Theorem K.6, Corollary K.7	Fixed bank, fixed positive dose, and the stated mean flow yield probability floors. Uniformity needs full coverage and equal categorical weights; unbanked modes may vanish.
Exemplar loss can bound key probability	Conditional lemma: Lemma K.8	A bounded joint potential and complete-response likelihood under the same sampling law protect those prompts. Neither held-out transfer nor visibility in eight draws follows.
Replay improves measured endpoints	Controlled comparisons: Figures 4 and 5	Paired replay additions isolate the implemented intervention within the reported model/domain cohorts. Partial 3B summaries are descriptive; individual-mode retention is not identified.
Equal key weights can aid breadth	Controlled weighting comparison: Appendix E.6	Uniform versus fresh-frequency weighting changes extra sampled modes in the reported cohorts; the average correctness contrast is inconclusive.
Replay can help on Level 2	Controlled within-level comparisons: Figure 6	Four complete domains support within-level replay effects. Separate terminal test splits prevent a pure cross-level difficulty-effect interpretation.
Fixed exemplars show heterogeneous score changes	Exploratory longitudinal finding: Appendix F.3	Central scores improve while a tail declines. No matched no-replay bank arm identifies a causal retention effect or exact key probabilities.
Accurate hosted outputs can be concentrated	Exploratory deployment comparison: Appendix H	Observed correctness and breadth describe the stated protocols. They do not identify RLVR as the cause, a model-size effect, or training-time collapse.
Temperature changes observed breadth	Exploratory matched-prompt comparison: Appendix H.3	Four temperature settings describe one deployment and reasoning profile. The best observed point does not establish an optimal sampling temperature in general.

Table 42: **What each central claim rests on.** Identities concern the sampling law, theorems specify conditional dynamics, and experiments support the stated intervention or exploratory observation. All reported intervals are nominal and unadjusted.

782 The before/after verifier-training precheck is a longitudinal checkpoint comparison (Appendix F.1).
783 Its extra-mode declines are measured changes in the registered runs, not proof of zero probability for
784 lost keys or a universal decrease in raw breadth across domains.

785 K Mode Collapse and Verified Replay

786 Under the explicit categorical assumptions below, centered binary-reward estimators amplify a pre-
787 existing imbalance among correct modes, whereas persistent verified replay supplies a loss barrier for
788 banked modes. The complete-response argument then states what would be needed to transfer that
789 protection to a language model. These results concern idealized update rules; they do not certify the
790 implemented AdamW/PPO trajectories.

791 K.1 Categorical abstraction and mean updates

792 Fix one isolated prompt and a finite category set $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, with $m \geq 2$ correct execution modes
 793 and $n \geq 1$ incorrect categories. Assign one independently optimized logit to each category, including
 794 one logit per canonical correct mode, and set $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$.
 795 Thus P measures correctness and q the conditional distribution among correct modes. Logits are
 796 finite initially, $0 < P(0) < 1$, and explicit reference KL is zero. The collapse result uses Euclidean
 797 infinitesimal expected updates in these logits, without optimizer preconditioning, momentum, or
 798 another regularizer.

799 A language model induces probabilities over canonical outcomes, but this identification of probabili-
 800 ties does not identify its update geometry. Aggregating several response strings into a mode does
 801 not generally commute with gradient flow in response logits. One independent logit per canonical
 802 outcome is therefore an explicit reference abstraction, not a reduction proved for the neural policy or its
 803 validator equivalence classes.

804 Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. At the
 805 on-policy point where the PPO ratio equals one, consider the common-length estimator

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (8)$$

806 For the Dr.GRPO abstraction, $w(M) = 1$ after absorbing the shared positive length scale. Reward-
 807 standardized GRPO with equal or common response-length normalization takes $w(M)$ to be the
 808 reciprocal group reward standard deviation on mixed groups (with a fixed numerical stabilizer if
 809 used). Standard variable-length GRPO also weights each response by its own inverse length [Shao
 810 et al., 2024, Liu et al., 2025]; that factor cannot generally be absorbed into $w(M)$ and is outside the
 811 equivalence proved here. In either included case, $0 < w(\ell) < \infty$ for $\ell \in \{1, \dots, G-1\}$. We define
 812 the entire update as zero on all-equal groups without evaluating an undefined reciprocal standard
 813 deviation; likewise, $w(M)M(G-M)$ below is defined as zero when $M = 0$ or $M = G$. All
 814 advantages and normalization weights are detached in the score estimator.

815 **Lemma K.1** (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the*
 816 *expected update in Equation (8) is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G-M)]}{G^2 P(1-P)} > 0. \quad (9)$$

817 *For the Dr.GRPO abstraction, $c_G(P) = (G-1)/G$. Group centering and reward standardization*
 818 *rescale the expected gradient without changing its direction under these assumptions.*

819 *Proof.* Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 0] = \nabla_z \log(1-P).$$

820 There are M correct rows with centered reward $(G-M)/G$ and $G-M$ incorrect rows with centered
 821 reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (8) is

$$w(M) \frac{M(G-M)}{G} (\nabla_z \log P - \nabla_z \log(1-P)) = w(M) \frac{M(G-M)}{GP(1-P)} \nabla_z P.$$

822 The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (9). For $w \equiv 1$, $\mathbb{E}[M(G-M)] = G(G-1)P(1-P)$, proving the Dr.GRPO value. $\square$

824 The binomial identity also gives the denominator-free expression

$$c_G(P) = \frac{G-1}{G} \sum_{j=0}^{G-2} \binom{G-2}{j} w(j+1) P^j (1-P)^{G-2-j}.$$

825 The binomial weights sum to one, so c_G lies between $(G-1) \min_j w(j+1)/G$ and $(G-1) \max_j w(j+1)/G$. This establishes the smooth, positive extension to $P = 0, 1$ used in the
 826 subsequent flow and convergence arguments.
 827

828 The next lemma specializes the practical-estimator analysis of Tajwar et al. [2026, Section 4.3 and
 829 Appendix D, Theorem 5] to our categorical notation; we include the derivation for completeness.

830 **Lemma K.2** (Binary MaxRL has the same correctness-gradient direction). *Under the same cate-*
831 *gorical and common-length assumptions, let MaxRL assign $A_i^{\text{MaxRL}} = GR_i/M - 1$ when $M > 0$*
832 *and zero when $M = 0$, as in Section 3. For $\hat{g}_G^{\text{MaxRL}} = G^{-1} \sum_i A_i^{\text{MaxRL}} \nabla_z \log p_{A_i}$ at the on-policy*
833 *point, the expected update is*

$$\mathbb{E}[\hat{g}_G^{\text{MaxRL}}] = c_G^{\text{MaxRL}}(P) \nabla_z P, \quad c_G^{\text{MaxRL}}(P) = \frac{1 - (1 - P)^{G-1}}{P} > 0, \quad (10)$$

834 *with continuous endpoint values $c_G^{\text{MaxRL}}(0) = G - 1$ and $c_G^{\text{MaxRL}}(1) = 1$.*

835 *Proof.* Condition on $M = m > 0$. Each successful row has advantage $(G - m)/m$ and each failed
836 row has advantage -1 . Using the same conditional score identities as above, the expected unaveraged
837 sum is

$$(G - m)(\nabla_z \log P - \nabla_z \log(1 - P)) = \frac{G - m}{P(1 - P)} \nabla_z P.$$

838 After the outer factor $1/G$ and averaging over $M \sim \text{Binomial}(G, P)$,

$$\mathbb{E}[(G - M)\mathbf{1}\{M > 0\}] = G(1 - P) - G \Pr(M = 0) = G[(1 - P) - (1 - P)^G].$$

839 Substitution gives Equation (10). Its endpoint values follow by continuity (or the finite geometric-
840 series identity), which also gives the stated continuous extension at both boundaries. $\square$

841 The same finite-series identity gives the potential used below:

$$\Psi_{\text{MaxRL}}(P) := \int_0^P c_G^{\text{MaxRL}}(s) ds = \sum_{k=1}^{G-1} \frac{1 - (1 - P)^k}{k}. \quad (11)$$

842 In particular, $\Psi_{\text{MaxRL}}(1) = \sum_{k=1}^{G-1} 1/k$. For binary rewards, $\text{best}@k$ equals $\text{pass}@k$, so this is
843 a harmonic mixture of sampling objectives for $k = 1, \dots, G - 1$. The upper limit is $G - 1$, not
844 G : the implemented centered estimator drops the entire update on all-failure groups. Subtracting
845 an unconditional score baseline even on those groups instead gives the order- G estimator. This
846 distinction concerns the exact finite-group expectation at the on-policy point; it does not assert
847 unbiasedness of subsequent clipped or adaptive-optimizer updates.

848 K.2 Winner-take-all collapse in the categorical mean flow

849 Consider $\dot{z} = c_G(P) \nabla_z P$, with any of the smooth positive coefficients established above: Dr.GRPO,
850 common-length reward-standardized GRPO, or the specified centered binary MaxRL estimator.
851 This infinitesimal expected-update model removes sampling noise and retains independently trained
852 Euclidean logits. The conditional dynamics reduce to the standard frequency-dependent replicator
853 system with fitness $f_c(q) = q_c$ [Hofbauer and Sigmund, 1998]. The result is a specialization to
854 the group estimators, not a new general theorem about replicator dynamics. Related correctness-
855 indifferent collapse analyses include Sinha et al. [2026], Lochab et al. [2026].

856 **Theorem K.3** (Standard replicator specialization for binary-reward RL). *Under the independent-*
857 *logit mean-flow assumptions above, start from finite logits with $0 < P(0) < 1$. If the conditional*
858 *correct-mode distribution $q(0)$ has a unique largest coordinate c^* , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

859 *Thus these categorical mean flows converge to a single correct execution mode for generic initial*
860 *conditions, even though every mode in $\mathcal{C}$ has exactly the same reward. This is an asymptotic statement*
861 *within the specified update geometry, not a claim of finite-time support loss or a convergence*
862 *guarantee for the stochastic neural optimizers used in our experiments.*

863 *Proof.* The coefficient c_G is smooth and bounded on $[0, 1]$, and the softmax reward gradient is
864 bounded. Thus the logit flow exists for all finite times and cannot develop infinite logits at finite time.
865 For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P) p_c(1 - P), \quad \dot{z}_a = -c_G(P) p_a P.$$

Also $\dot{P} = c_G(P) \|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G , together with

$$\|\nabla_z P\|_2^2 \geq P^2(1-P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s)) P(s) (1 - P(s)) ds.$$

If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$. Therefore $\tau(t) \rightarrow \infty$, leaving infinite effective time for the correct-simplex dynamics below.

On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (12)$$

The initially unique maximum remains the unique maximum. For $c \neq c^*$, set $\xi_c = q_c/q_{c^*}$, so $\xi_c(0) < 1$ and ξ_c is nonincreasing. Since $q_{c^*} \geq 1/m$,

$$\frac{d \log \xi_c}{d\tau} = -q_{c^*} (1 - \xi_c) \leq -\frac{1 - \xi_c(0)}{m}.$$

Hence $\xi_c(\tau) \leq \xi_c(0) \exp[-(1 - \xi_c(0))\tau/m] \rightarrow 0$. Every competing mode vanishes relative to c^* ; normalization then yields $q_{c^*} \rightarrow 1$, establishing the claimed single-mode limit. $\square$

If several modes tie for the largest initial probability, symmetry preserves the tie; the same ratio argument eliminates strictly smaller coordinates and leaves the tied maxima uniform. An exactly uniform correct distribution is therefore a special initial condition. The theorem concerns limiting concentration: finite softmax logits already give positive probabilities at every finite time, so finite-time positivity alone is not a retention result.

Dependence on optimization geometry. For a smooth parameterization $z = z(\theta)$ with Jacobian $J_\theta = \partial z / \partial \theta$, Euclidean parameter flow instead induces $\dot{z} = c_G(P) J_\theta J_\theta^\top \nabla_z P$. The independently trained logit theorem uses $J_\theta J_\theta^\top = I$ and does not automatically extend to arbitrary shared parameters or another optimization metric. Equal binary reward alone does not force this winner selection: the categorical natural-gradient reward flow can be represented by $\dot{z}_a = c_G(P)(R_a - P)$, which gives identical velocities to correct logits and preserves their ratios. Here $R_a = \mathbf{1}[a \in \mathcal{C}]$. Parameter count alone does not specify the optimization geometry. The empirical scale comparisons also change pretrained models and learning-rate schedules; they neither measure the geometry nor identify the cause of cross-scale differences. The theorem supplies a mechanism to test in controlled training; it cannot identify the training history or cause of concentration in a hosted model.

K.3 A replay signal when fresh groups supply none

Lemma K.4 (Fresh-group starvation and replay signal). *Fix the categorical policy above with correctness probability P , and draw $G \geq 2$ fresh categories independently. The probability of a mixed-reward group is*

$$h_G(P) = 1 - P^G - (1 - P)^G \leq G \min\{P, 1 - P\}.$$

Every all-equal group has zero fresh task component for Equation (8) and the specified centered MaxRL estimator. A nonzero fresh task component therefore requires a mixed group, so its probability is at most h_G . This gives no lower bound on gradient magnitude and says nothing about other loss terms, clipping, or an optimizer step carrying momentum across successive updates.

For a nonempty fixed verified bank $\mathcal{B} \subseteq \mathcal{C}$ with normalized positive weights w_b , let $R_w = -\sum_b w_b \log p_b$ and $\rho_w > 0$. Extend w by zero outside the bank to obtain $\bar{w}$. The deterministic replay ascent vector is

$$v^{\text{replay}} = -\rho_w \nabla R_w = \rho_w (\bar{w} - p),$$

904 which vanishes only at $p = \bar{w}$. In particular, if $p_b \leq w_b/2$ for a banked mode, then $v_b^{\text{replay}} \geq \rho_w w_b/2$,
 905 even when b is absent from the fresh group. For a uniform bank of $k \geq 2$ modes and fixed $\rho_w > 0$,
 906 as $p \rightarrow \mathbf{e}_b$ at one banked correct mode, $h_G(P) \rightarrow 0$ whereas $\|v^{\text{replay}}\|_2 \rightarrow \rho_w \sqrt{1 - 1/k} > 0$. The
 907 replay norm therefore stays positive as fresh task-signal opportunities vanish.

908 *Proof.* The all-correct and all-incorrect events are disjoint and have probabilities P^G and $(1 - P)^G$.
 909 A mixed group requires both a correct and an incorrect sample; the union bound gives the two bounds
 910 GP and $G(1 - P)$. For $R_w = -\sum_b w_b \log p_b$, the softmax score identity gives $\partial R_w / \partial z_a = p_a - \bar{w}_a$,
 911 proving the replay formula and coordinate bound. Finally,

$$\|\bar{w} - \mathbf{e}_b\|_2^2 = (1 - 1/k)^2 + (k - 1)/k^2 = 1 - 1/k,$$

912 which gives the strictly positive limiting norm for $k \geq 2$. $\square$

913 The displayed replay vector averages the fixed bank. A stochastic exemplar draw has this vector only
 914 in expectation, and a nonzero logit coordinate can be annihilated by a degenerate neural Jacobian.
 915 Replay also need not increase the next mixed-group probability: $h_G(P)$ is largest at $P = 1/2$. The
 916 claim is that a banked target can supply a separate verified signal when the fresh task gradient is
 917 unavailable.

918 A bounded advantage based on fresh samples has a different boundary behavior. The next bound
 919 includes group-dependent detached scores, and distinguishes that estimator from exact entropy
 920 regularization.

921 **Lemma K.5** (Bounded sampled scores have no probability-independent boundary force). *In the*
 922 *same independent-logit model, condition on the training history, draw $A_1, \dots, A_G$ independently*
 923 *from p , and let a_i be any detached, possibly group-dependent advantages with $|a_i| \leq B$. To interpret*
 924 *the bound uniformly as $p_b \rightarrow 0$, assume a finite B independent of that limit. At the on-policy point*
 925 *define*

$$\hat{g} = \frac{1}{G} \sum_{i=1}^G a_i (\mathbf{e}_{A_i} - p).$$

926 For every category b ,

$$|\mathbb{E}[\hat{g}_b]| \leq \mathbb{E}|\hat{g}_b| \leq 2Bp_b(1 - p_b).$$

927 On a group with no draw of b , exactly $\hat{g}_b = -p_b G^{-1} \sum_i a_i$; it is zero if the applied advantages
 928 sum to zero. By comparison, the verified-replay component for a fixed target w has logit velocity
 929 $\rho(w_b - p_b)$, which approaches $\rho w_b > 0$ for a banked category as $p_b \rightarrow 0$.

930 *Proof.* Use the triangle inequality and $|a_i| \leq B$. Each marginal draw satisfies $\mathbb{E}|\mathbf{1}\{A_i = b\} - p_b| =$
 931 $2p_b(1 - p_b)$, irrespective of the dependence of a_i on the group. On the absence event all indicator
 932 terms are zero. The replay formula follows by differentiating $-\sum_b w_b \log p_b$. $\square$

933 The bound compares the isolated logit-gradient components. It does not prove inevitable collapse,
 934 rule out retention by a bounded estimator, or guarantee that replay dominates the full update. An
 935 unsampled mode can still change through softmax normalization or shared parameters. Exact entropy
 936 is outside the uniformly bounded-score premise because its surprisal is unbounded near zero; the
 937 lemma is not an impossibility result for entropy regularization. Nor does the absence of a mode from
 938 a finite sample establish zero probability.

939 K.4 Fixed-bank replay and qualitative retention

940 Let $\mathcal{B} \subseteq \mathcal{C}$ be a nonempty fixed bank of k modes that the policy has produced and the validator
 941 has accepted. Uniform verified replay uses one exemplar per banked mode and, in the categorical
 942 reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (13)$$

943 This is ordinary cross-entropy against the distribution that assigns mass $1/k$ to each banked mode
 944 and zero elsewhere. It is not conditioned on the bank: decreasing (13) can raise total bank mass as
 945 well as redistribute mass within the bank.

946 In an averaged continuous-time model, absorb a fixed round-robin replay frequency and loss scale
 947 into an effective dose $\rho > 0$. This assumes a fixed eligible-bank set and constant relative weighting;
 948 it does not identify finite alternating updates with simultaneous gradient flow. The prompt-isolated
 949 categorical mean flow for any of the three fresh binary-reward objectives above plus verified replay is

$$\dot{z} = c_G(P)\nabla_z P - \rho\nabla_z R_{\mathcal{B}}(z). \quad (14)$$

950 Let $\Psi(P) = \int_0^P c_G(s) ds$. For the finite-group estimators in Lemmas K.1 and K.2, c_G has a
 951 finite positive extension to $[0, 1]$; in particular, Dr.GRPO has $c_G = (G - 1)/G$ and MaxRL has
 952 $c_G = [1 - (1 - P)^{G-1}]/P$. Define

$$F_{\mathcal{B}}(z) = \rho R_{\mathcal{B}}(z) - \Psi(P(z)). \quad (15)$$

953 Then (14) is exactly gradient flow $\dot{z} = -\nabla_z F_{\mathcal{B}}$.

954 **Theorem K.6** (Verified replay prevents banked-mode extinction). *Suppose logits are finite at time T*
 955 *and thereafter the nonempty verified bank $\mathcal{B}$ is fixed, fits within replay capacity, and receives a fixed*
 956 *positive effective replay dose ρ . Under (14), define*

$$C_T = R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho}.$$

957 Then for every $b \in \mathcal{B}$ and every $t \geq T$,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (16)$$

958 Thus each banked probability is uniformly bounded away from zero for all $t \geq T$, including
 959 asymptotically, even when the bank does not cover every correct mode. This is stronger than
 960 finite-time positivity, which already follows from finite softmax logits without replay.

961 *Proof.* The smooth vector field has bounded velocity because c_G , $\nabla_z P$, and $\nabla_z R_{\mathcal{B}}$ are bounded, so
 962 finite initial logits give a flow for every finite $t \geq T$. Along this flow,

$$\frac{d}{dt} F_{\mathcal{B}}(z(t)) = -\|\nabla_z F_{\mathcal{B}}(z(t))\|_2^2 \leq 0.$$

963 Since $P(t) \leq 1$ and Ψ is increasing,

$$R_{\mathcal{B}}(z(t)) \leq R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho} = C_T.$$

964 Every $-\log p_b$ is nonnegative and their sum is $kR_{\mathcal{B}}$. Therefore $-\log p_b(t) \leq kC_T$ for every banked
 965 mode, which is exactly Equation (16) for every time after T . $\square$

966 The floor is qualitative and can be numerically vacuous. For example, at $k = 16$ even a hypothetical
 967 $C_T = 160$ gives e^{-2560} , far below any useful evaluation probability. The argument establishes a
 968 uniform-in-time lower bound in the stated ideal flow, not practical visibility or a prediction for the
 969 experimental replay loss. The effective dose also depends on relative fresh/replay frequencies, not
 970 the replay coefficient alone.

971 A common nonnegative, locally integrable rate $\eta(t)$ multiplying both terms of (14) preserves the
 972 energy bound, since $\dot{F}_{\mathcal{B}} = -\eta(t)\|\nabla F_{\mathcal{B}}\|_2^2 \leq 0$. The convergence statements below additionally
 973 require $\int_T^\infty \eta(t) dt = \infty$ and then follow by reparameterizing optimization time. Infinitely many
 974 replay visits alone establish neither this condition nor a fixed replay-to-fresh weighting. Different
 975 time-varying relative weights are outside the fixed-potential argument.

976 **Full coverage and the categorical target.**

977 **Corollary K.7** (Conditional full-coverage categorical limit). *Assume the conditions of Theorem K.6*
 978 *and that by time T the bank covers the full correct support, $\mathcal{B} = \mathcal{C}$. Let $u_c = 1/m$ on $\mathcal{C}$. Then the*
 979 *categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

980 Consequently

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

981 and conditional correct-mode collapse is impossible.

982 *Proof.* For full coverage,

$$R_C = \log m - \log P + \text{KL}(u||q). \quad (17)$$

983 The potential is bounded below by $-\Psi(1)$, so its descent gives $\int_T^\infty \|\nabla F_C\|_2^2 dt < \infty$. The finite-
 984 group c_G is smooth on $[0, 1]$, and softmax derivatives and the cross-entropy gradient are bounded.
 985 Thus the Hessian of F_C and the velocity are bounded; the gradient along the trajectory is uniformly
 986 continuous. Square integrability then implies $\nabla F_C \rightarrow 0$: recurring gradients of fixed positive size
 987 would otherwise contribute energy over intervals of a uniform minimum length.

988 Every limiting probability vector therefore satisfies the correct-coordinate equations

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P).$$

989 Summing them gives $0 = -(1 - P)(\rho + c_G(P)P)$, hence $P = 1$. Each correct coordinate is then
 990 $1/m$ and every incorrect coordinate is zero. Compactness of the probability simplex and uniqueness
 991 of this limit establish convergence of the probability vector. $\square$

992 **Positive weights and the implemented length normalization.** Uniformity of the target is separate
 993 from protection against extinction. For fixed weights $w_b > 0$ with $\sum_{b \in \mathcal{B}} w_b = 1$, define $R_w =$
 994 $-\sum_b w_b \log p_b$ and use replay coefficient $\rho_w > 0$. The same potential argument gives, for all $t \geq T$,

$$p_b(t) \geq \exp(-C_{T,w}/w_b), \quad C_{T,w} = R_w(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

995 If $\mathcal{B} = \mathcal{C}$, then $R_w = H(w) - \log P + \text{KL}(w||q)$, and the independent-logit limit is $P \rightarrow 1$, $q \rightarrow w$.
 996 The preceding stationarity proof applies with $1/m$ replaced by w_c : summing the correct-coordinate
 997 equations forces $P = 1$, then each equation forces $p_c = w_c$.

998 A partial bank has a stronger limitation in this same ideal flow. For any fixed nonempty $\mathcal{B} \subseteq \mathcal{C}$ and
 999 fixed positive weights summing to one, extend w by zero to $\bar{w}$ on all categories. Then $p(t) \rightarrow \bar{w}$,
 1000 so even correct modes outside the frozen bank vanish asymptotically. For $F = \rho_w R_w - \Psi(P)$, the
 1001 preceding energy and bounded-Hessian argument still gives $\nabla F \rightarrow 0$; neither argument requires full
 1002 bank coverage. At every subsequential probability limit, the correct-coordinate equations are

$$0 = \rho_w(p_c - \bar{w}_c) - c_G(P)p_c(1 - P).$$

1003 Summing over correct categories gives $0 = -(1 - P)(\rho_w + c_G(P)P)$, hence $P = 1$; each equation
 1004 then gives $p_c = \bar{w}_c$, and all incorrect mass is zero. Compactness and uniqueness of this limiting
 1005 probability vector prove convergence. Thus protecting a fixed set of banked modes need not preserve
 1006 unbanked correct alternatives. This conclusion concerns the fixed-bank categorical flow, not changing-
 1007 bank or stochastic neural training.

1008 This distinction matters for the implemented mean-token exemplar loss. Even if each key were
 1009 represented by exactly one categorical exemplar of fixed length L_b , its loss would be AR_w , where

$$A = \frac{1}{k} \sum_{b \in \mathcal{B}} \frac{1}{L_b}, \quad w_b = \frac{L_b^{-1}}{\sum_{j \in \mathcal{B}} L_j^{-1}}.$$

1010 Its effective categorical replay coefficient would be $\rho_w = \rho_{LM}A$. The full-coverage limit would
 1011 therefore weight shorter exemplars more, and would be uniform only for equal lengths. For an actual
 1012 language model, $\pi_\theta(e_b | x)$ is only one contribution to the total key probability. Equal averaging
 1013 of per-token exemplar scores does not imply uniform learned key probabilities; the categorical
 1014 uniformity corollary is not that claim.

1015 Full support is an additional premise, not a consequence of replay. In Python factors, prompts admit
 1016 16–3,600 correct execution modes (mean 229.44), while the bank capacity is 16. The full-coverage
 1017 corollary therefore cannot supply a domain-wide guarantee there. The retention theorem protects
 1018 only admitted banked modes; it supplies no probability floor for an unbanked key.

1019 K.5 Complete responses and finite evaluation visibility

1020 The learner scores a particular response string, while evaluation groups strings by execution key.
 1021 Event containment provides a conditional bridge: one complete verified response is one way of
 1022 producing its key. This bridge also makes the loss's length factor explicit. Here $p_\theta(b | x) =$
 1023 $\Pr_{Y \sim \pi_\theta(\cdot | x)}[V(Y, s_x) = b]$ is the unconditional probability of producing key b , rather than the
 1024 probability conditional on correctness.

1025 **Lemma K.8** (Retention from a joint exemplar potential). *Fix finitely many verified complete-response*
 1026 *exemplars (x_j, e_j) , each realizing key b_j , with lengths $L_j > 0$ and fixed weights $\alpha_j > 0$. Let*

$$F(\theta) = \sum_j \alpha_j \frac{-\log \pi_\theta(e_j | x_j)}{L_j} - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

1027 *Suppose $F(\theta(T))$ is finite and $F(\theta(t)) \leq F(\theta(T))$ for every $t \geq T$. With $D = F(\theta(T)) + J_{\max}$,*
 1028 *every protected exemplar satisfies*

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp\left(-\frac{L_j D}{\alpha_j}\right) > 0 \quad (t \geq T).$$

1029 *Proof.* The weighted exemplar loss equals $F(\theta(t)) + J(\theta(t)) \leq D$. Each summand is nonnegative,
 1030 so $-\log \pi_\theta(e_j | x_j) \leq L_j D / \alpha_j$. The complete-response event is a subset of the event producing key
 1031 b_j , which proves the stated lower bound on every protected key’s probability. $\square$

1032 Exact descent of a smooth joint potential is sufficient for the energy inequality, even with shared
 1033 parameters and multiple prompts. For example, the ideal categorical model permits $J(\theta) =$
 1034 $\sum_x \nu_x \Psi_x(P_x(\theta))$ with fixed nonnegative prompt weights and $J_{\max} = \sum_x \nu_x \Psi_x(1)$. The lemma
 1035 does not establish this descent condition for stochastic, clipped, alternating AdamW updates, and
 1036 it neither implies uniform neural key probabilities nor requires them. Its bound applies only to the
 1037 listed exemplar–prompt pairs. Shared parameters do not extend it to held-out prompts; transfer to
 1038 those prompts remains an empirical question.

1039 The scored probability must describe the complete response, including its termination event or a fixed
 1040 deterministic horizon, under the same prompt, temperature, masks, truncation rules, and verifier as
 1041 the sampling law being bounded. A prefix score or a score under another decoding law does not
 1042 supply this bound without a separate event or likelihood comparison. For a complete response of
 1043 length L and mean-token log score s , its probability is e^{Ls} , not e^s . A change in this exemplar’s
 1044 likelihood need not equal the change in its whole key’s probability.

1045 **Retention and finite evaluation visibility.** At a fixed checkpoint, suppose k distinct banked keys
 1046 for one prompt have probability floors $p_b \geq \delta_b > 0$. For K independent draws from that same policy,
 1047 the banked distinct count $D_{\mathcal{B}, K}$ satisfies

$$\mathbb{E}[D_{\mathcal{B}, K}] \geq \sum_{b \in \mathcal{B}} [1 - (1 - \delta_b)^K],$$

$$\Pr(\text{some banked key is unseen}) \leq \sum_{b \in \mathcal{B}} (1 - \delta_b)^K \leq k e^{-K \delta_{\min}},$$

1048 where $\delta_{\min} = \min_b \delta_b$. The first statement follows from indicator expectations and the second from
 1049 the union bound. Thus $K \geq \log(k/\varepsilon)/\delta_{\min}$ suffices to observe all banked keys with probability at
 1050 least $1 - \varepsilon$, for $0 < \varepsilon < 1$. The energy floors can make this budget impractically large; asymptotic
 1051 retention does not imply visibility in an eight-draw evaluation.

1052 Conversely, not observing a mode does not establish extinction. At a fixed checkpoint, zero sightings
 1053 of a prespecified key in N independent draws give a one-sided $(1 - \alpha)$ binomial upper bound $1 - \alpha^{1/N}$
 1054 and lower bound zero [Clopper and Pearson, 1934]. For $N = 32$ and $\alpha = .05$, the upper bound is
 1055 approximately 8.94%. These are single-key bounds; simultaneous statements need error allocation.
 1056 Counts from changing checkpoints cannot be pooled to certify a current policy, and teacher-forced
 1057 scores are not independent Bernoulli observations.

1058 K.6 Scope relative to entropy, admission, and neural optimization

1059 **Exact entropy is a different comparator.** For a known full correct support and uniform u , the
 1060 elementary identities

$$H(q) = \log m - \text{KL}(q \| u), \quad R_u(p) = \log m - \log P + \text{KL}(u \| q)$$

1061 show that exact conditional entropy and categorical replay both favor a uniform conditional target.
 1062 With strictly increasing correctness potential Ψ , both $\Psi(P) + \beta H(q)$ and $\Psi(P) - \rho R_u(p)$ have the

unique global optimum $P = 1, q = u$ in probability space on the simplex with $P > 0$, for positive coefficients. This is a boundary limit for finite-logit softmax policies. The optimum follows from nonnegativity of KL and increasing Ψ ; it does not assert convergence of either neural update. The KL direction distinguishes their loss level sets [Pereyra et al., 2017, Agarwal et al., 2021]: replay loss diverges when a protected coordinate vanishes, whereas conditional entropy can remain $\log(m - 1)$ with one missing mode.

Exact full-output entropy is a third objective and also regularizes incorrect mass. Exact tabular entropy-regularized policy gradient has its own positive probability and convergence guarantees under specified assumptions [Mei et al., 2020]; our collapse theorem contains no entropy term. The sampled semantic scores in the experiments are not exact gradients of full-output or conditional verified-key entropy. Neither their empirical comparison nor the bounded-score lemma is an impossibility result for entropy.

Discovery and optimization remain separate requirements. Retention starts after a verified key enters active memory. Actual insertion depends on generation, validation, capacity, and eligibility; a positive sampling probability alone does not establish eventual admission. Once a non-evicting capacity-16 bank fills, an unbanked key can remain ineligible. Changing membership or exemplar weights changes the loss being bounded, so fixed-bank descent cannot simply be carried across those changes. In particular, the fixed-weight argument does not cover a target whose weights continue to change with fresh observation counts.

For a fixed joint potential, a discrete bound $F(\theta_{n+1}) \leq F(\theta_n) + \epsilon_n$ with $\epsilon_n \geq 0$ and $\sum_{n \geq n_0} \epsilon_n = E < \infty$ would extend Lemma K.8, using $D = F(\theta_{n_0}) + J_{\max} + E$ for steps $n \geq n_0$. This follows by telescoping and bounds retention, not convergence. The noisy optimizer would need a separate argument establishing such control of the joint loss’s upward drift. A positive replay coefficient or small nominal learning rate does not establish it. Adam moments, clipping, momentum, prompt scheduling, and bank changes are not covered by the exact-flow proofs. The experiments have not established the required joint-energy or update-error assumptions. The mathematical contribution is the stated categorical mechanism and conditional exemplar argument; empirical support and longitudinal exemplar scores remain separate measurements.

L Source Integrity and Reproducibility

L.1 Admissible sources and missing checkpoints

Falcon3-1B Countdown ReplayDr.GRPO seed 59 contains two complete four-draw terminal evaluation sequences with identical request identities but conflicting metrics. The integrity amendment excludes that source at every checkpoint and from every efficacy endpoint, paired contrast, and aggregate that reuses it; neither sequence is selected or averaged. The independently valid Dr.GRPO and GRPO seed-59 sources remain admissible. The core replay comparison therefore has 74 pairs: 14 complete five-seed blocks and Falcon Countdown at $n = 4$ (seeds 55–58). Primary partial intersections are descriptive and receive no confidence interval.

Trajectory records use registered sources and approved continuations or replacements, with superseded attempts excluded. Identical duplicates are deduplicated; conflicting records invalidate the affected checkpoint rather than being selected by metric value. Missing or ambiguous checkpoints remain gaps; curves do not interpolate or carry values forward across them. An ambiguous nonterminal checkpoint does not invalidate an independently admissible terminal endpoint. Each curve declares its fixed source-admissible seed cohort; paired core curves use the admitted intersection. Accuracy and mode-count views use the same populations and gap rules. Supporting methods retain their explicit method-specific cohorts; available checkpoint counts and gaps are recorded alongside the curves. Machine-readable records identify the sources, exclusions, exact seed sets, checkpoint rule, and source hashes.

L.2 Data, models, and scientific records

The ModeBench dataset provides separate level/domain configurations, fixed split roles, full row identities, and construction and admission records. Its loading guide and evaluation guide identify the reported splits and preserve auxiliary data separately. The model archive links each available

1114 model–domain–method–seed export to an immutable upload revision containing original weights,
1115 configuration, tokenizer, file hashes, completion provenance, and the applicable base-model license.
1116 The experiment index maps the core scale comparisons, MaxRL factorial, Level-2 factorial, replay-
1117 weighting ablation, and supporting controls to their model families; the scientific-record index and
1118 restore guide identify preserved analysis inputs and SHA-256 verification procedures.

1119 Each empirical figure has a machine-readable record identified in the figure manifest, containing
1120 its displayed values and frozen numerical inputs. `paper/FIGURE_MANIFEST.md` maps both paper
1121 versions to their builders, records, and reproduction commands. Detailed source and recovery
1122 inventories remain in those records and the repository’s archived audit documents. Model availability
1123 may extend beyond the frozen analysis population; it does not change a figure’s source-admission
1124 or seed-intersection rule. Terminal exports restore inference models, while intermediate trajectories
1125 require the recorded evaluation checkpoints.

1126 **L.3 Exact continuation and validation**

1127 Exact training continuation requires the atomic optimizer, bank, token, cursor, and request-stream
1128 state in Appendix D.4, in addition to model weights. State mismatch, non-finite values, missing
1129 completion evidence, changed dataset identities, and source or aggregate hash drift are hard failures.
1130 Placement-only recovery may change hardware or wall time but must preserve the scientific update.
1131 Figure reproduction uses frozen records; missing results are never replaced by shallower checkpoints
1132 or silently incorporated from a newer archive.

1133 **M Limitations and Disclosures**

1134 The categorical theory depends on its stated geometry, energy, and coverage assumptions; it does not
1135 certify the implemented neural optimizer. Uniform correct mass requires full verified coverage, which
1136 Python’s capacity-16 bank generally precludes. The benchmark measures task-defined executable
1137 support on synthetic tasks, not free-form reasoning diversity or downstream utility. Experiments
1138 fix $G = 16$, zero reference KL, no token-entropy bonus, and one decoding protocol. The current
1139 results state every partial seed intersection and its uncertainty limits; intervals condition on the frozen
1140 prompts and evaluation requests.

1141 **M.1 AI Use Statement**

1142 Generative AI assisted manuscript preparation, mathematical analysis, and supporting code. The
1143 authors reviewed and verified the resulting content and take responsibility for the final manuscript.

1144 **M.2 Ethics Statement**

1145 The benchmark and experiment-specific data are synthetic; this study collects no human-subject or
1146 personal data. More verified modes are not inherently better: a distinct execution can be costly or
1147 risky despite passing a validator. Applications should report correctness, invalidity, and relevant costs
1148 alongside mode counts.

1149 **M.3 Reproducibility Statement**

1150 Claims use versioned results and fixed source-admission rules. Missing required results and aggregate
1151 drift cause validation to fail. Appendix B specifies split identities and evaluation settings; Appendix L
1152 links frozen scientific records, model exports, and reproduction procedures.